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RNNs for Sequence and Time-series Data

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Ph.D. in Computer Science

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code&data

Objectives

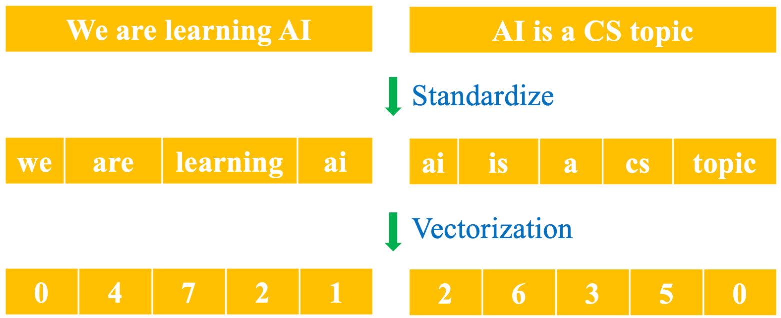
Text Classification

sample1: ‘We are learning AI’

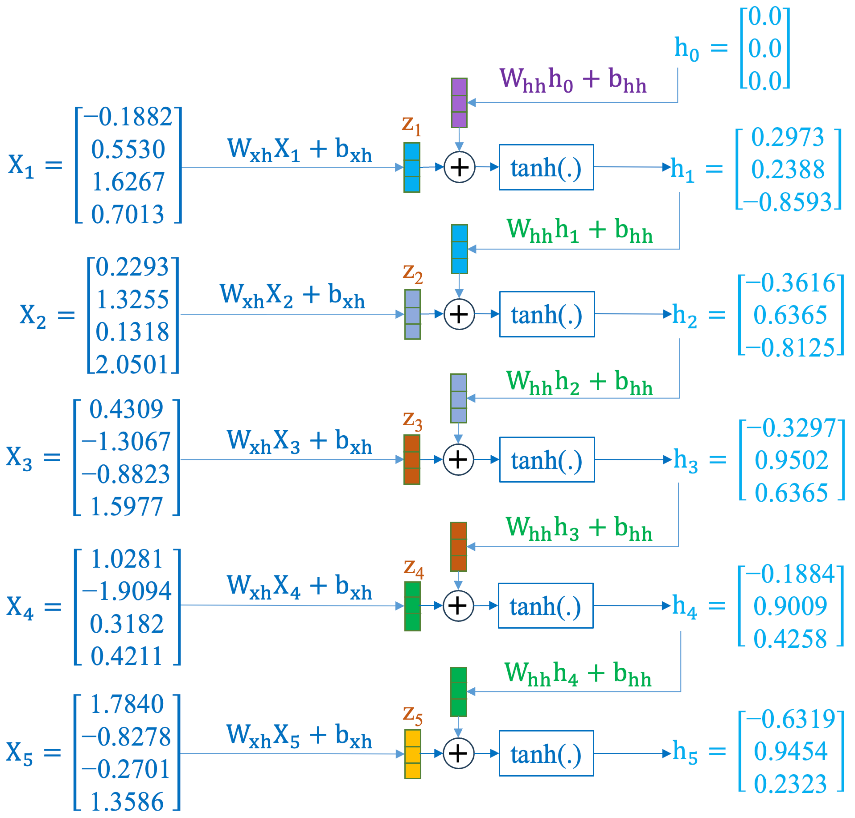
sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

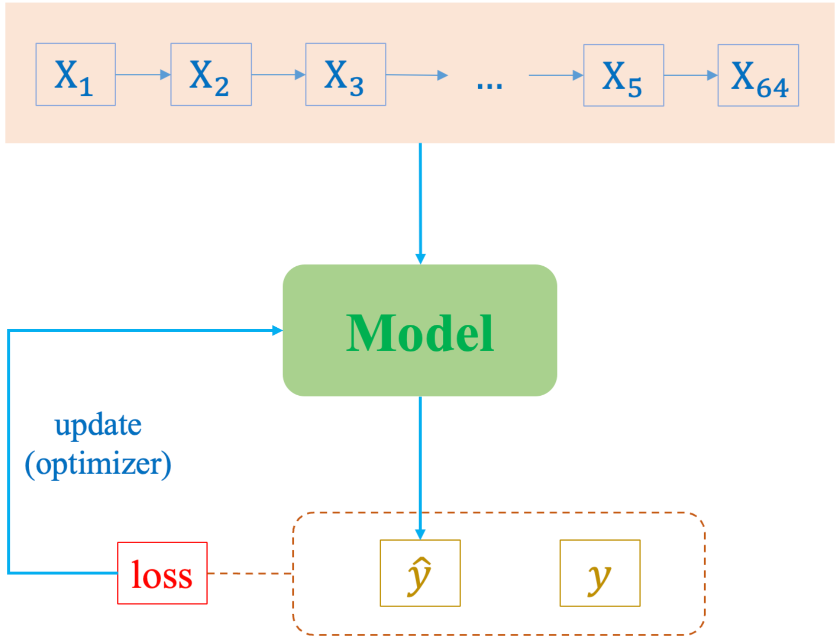
(2) Transform text into features



RNN Construction



RNNs for Time-series



Outline

SECTION 1

Text Classification

SECTION 2

RNN Construction

SECTION 3

RNNs for Time-series

sample1: 'We are learning AI'

sample2: 'AI is a CS topic'

(1) Build vocabulary from corpus

(2) Transform text into features

We are learning AI					AI is a CS topic				

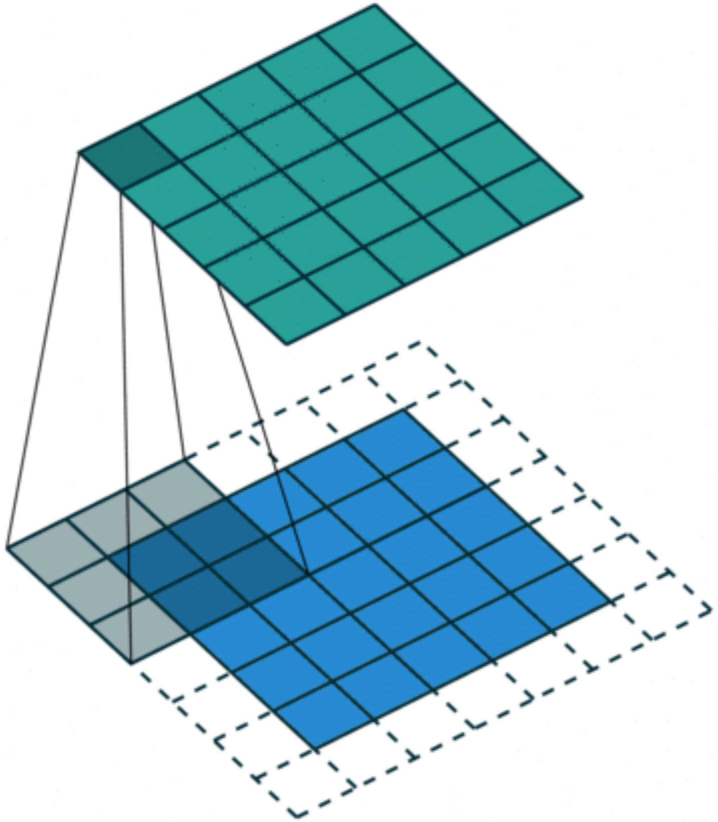
Standardize

we	are	learning	ai		ai	is	a	cs	topic

Vectorization

0	4	7	2	1	2	6	3	5	0

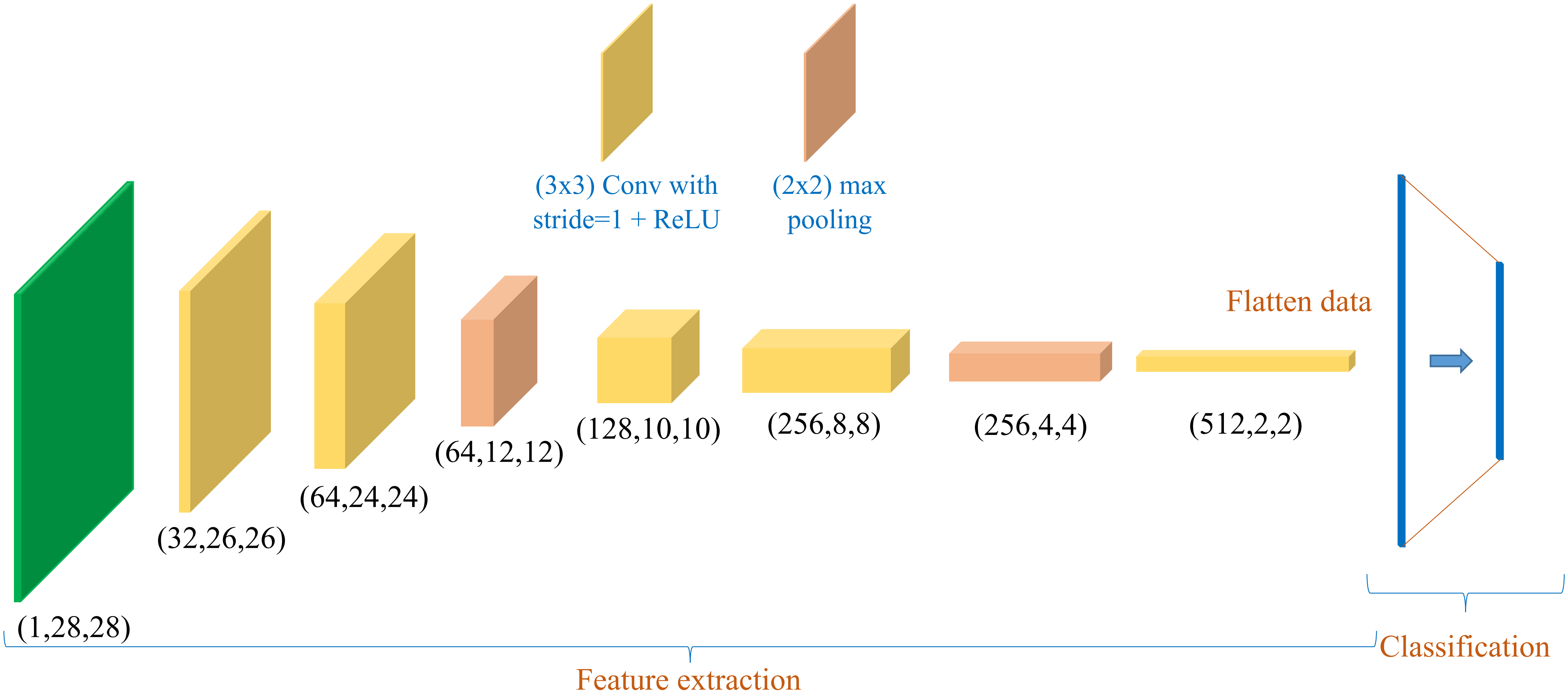
Image Data




airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck

Image Data

❖ Example



Text Classification

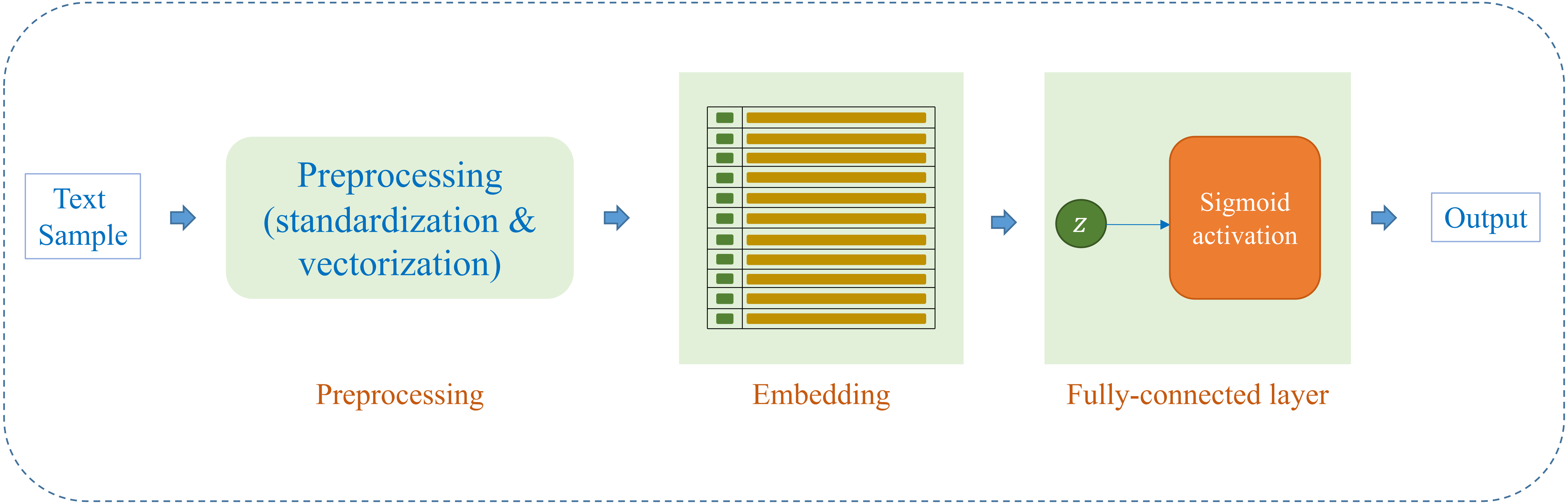
❖ IMDB dataset

- 50,000 movie review for sentiment analysis
- Consist of: + 25,000 movie review for training
  + 25,000 movie review for testing
- Label: positive – negative

"A wonderful little production. The filming technique is very unassuming- very old-time-BBC fashion and gives a comforting, and sometimes discomforting, sense of realism to the entire piece….."	positive
"This show was an amazing, fresh & innovative idea in the 70's when it first aired. The first 7 or 8 years were brilliant, but things dropped off after that. By 1990, the show was not really funny anymore, and it's continued its decline further to the complete waste of time it is today…."	negative
"I thought this was a wonderful way to spend time on a too hot summer weekend, sitting in the air conditioned theater and watching a light-hearted comedy. The plot is simplistic, but the dialogue is witty and the characters are likable (even the well bread suspected serial killer)…."	positive
"BTW Carver gets a very annoying sidekick who makes you wanna shoot him the first three minutes he's on screen."	negative

Text Classification

❖ Simple approach



Embedding

index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus

  sample1: ‘We are learning AI’

  sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

We are learning AI	AI is a CS topic

Standardize

we	are	learning	ai

ai	is	a	cs	topic

```python
from torchtext.data.utils import get_tokenizer

sample1 = 'We are learning AI'
sample2 = 'AI is a CS topic'

# Define tokenizer function
tokenizer = get_tokenizer('basic_english')
sample1_tokens = tokenizer(sample1)
sample2_tokens = tokenizer(sample2)

print(sample1_tokens)
print(sample2_tokens)
```

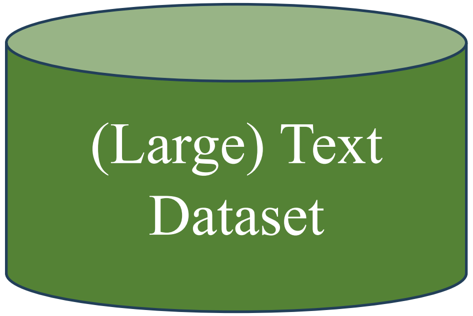
```
['we', 'are', 'learning', 'ai']
['ai', 'is', 'a', 'cs', 'topic']
```

Embedding

index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus
  - sample1: ‘We are learning AI’
  - sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus




```python
# problem?
from torchtext.data.utils import get_tokenizer
from torchtext.vocab import build_vocab_from_iterator

sample1 = 'We are learning AI'
sample2 = 'AI is a CS topic'
data = [sample1, sample2, ...]

# Create vocabulary
vocab = build_vocab_from_iterator(data)
```

#different words are enormous

How to represent ‘text’ effectively?

Embedding

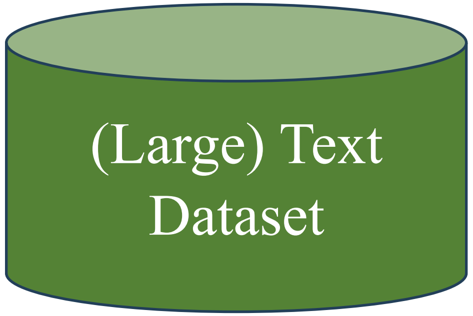
index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus

  sample1: ‘We are learning AI’

  sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus




```
from torchtext.data.utils import get_tokenizer
from torchtext.vocab import build_vocab_from_iterator

sample1 = 'We are learning AI'
sample2 = 'AI is a CS topic'
data = [sample1, sample2, ...]

# Create vocabulary
vocab_size = 8
vocab = build_vocab_from_iterator(data,
                                  max_tokens=vocab_size,
                                  specials=["<unk>",
                                            "<pad>"])
vocab.set_default_index(vocab["<unk>"])
```

#different words are enormous

How to represent ‘text’ effectively?

→ Use a limited number of words

Embedding

index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus

  sample1: ‘We are learning AI’

  sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

```
from torchtext.data.utils import get_tokenizer
from torchtext.vocab import build_vocab_from_iterator

sample1 = 'We are learning AI'
sample2 = 'AI is a CS topic'
data = [sample1, sample2]

# Create a function to yield list of tokens
tokenizer = get_tokenizer('basic_english')
def yield_tokens(examples):
    for text in examples:
        yield tokenizer(text)

# Create vocabulary
vocab_size = 8
vocab = build_vocab_from_iterator(yield_tokens(data),
                                  max_tokens=vocab_size,
                                  specials=["<unk>",
                                            "<pad>"])

vocab.set_default_index(vocab["<unk>"])
```

```
vocab.get_stoi()
```

```
{'<unk>': 0,
 '<pad>': 1,
 'ai': 2,
 'a': 3,
 'is': 6,
 'are': 4,
 'learning': 7,
 'cs': 5}
```

#different words are enormous

How to represent ‘text’ effectively?

➡ Use a limited number of words

➡ Get data sample-by-sample

Embedding

index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus
  sample1: 'We are learning AI'
  sample2: 'AI is a CS topic'

(1) Build vocabulary from corpus

(2) Transform text into features

We are learning AI					AI is a CS topic				
we	are	learning	ai		ai	is	a	cs	topic
0	4	7	2	1	2	6	3	5	0

Standardize

Vectorization

'We' 'are' 'learning' 'AI'

Demo

```
tokens = tokenizer(sample1)
print(tokens)

sample1_tokens = [vocab[token] for token in tokens]
print(sample1_tokens)
```

```
['we', 'are', 'learning', 'ai']
[0, 4, 7, 2]
```

```
tokens = tokenizer(sample2)
print(tokens)

sample2_tokens = [vocab[token] for token in tokens]
print(sample2_tokens)
```

```
['ai', 'is', 'a', 'cs', 'topic']
[2, 6, 3, 5, 0]
```

Embedding

index	0	1	2	3	4	5	6	7
word	[UNK]	pad	ai	a	are	cs	is	learning

- Example corpus
  - sample1: 'We are learning AI'
  - sample2: 'AI is a CS topic'

(1) Build vocabulary from corpus

(2) Transform text into features

We are learning AI	AI is a CS topic

Standardize

we	are	learning	ai

ai	is	a	cs	topic

Vectorization

0	4	7	2	1

2	6	3	5	0

sample3 = AI topic in CS is difficult'

```python
def vectorize(text, vocab, sequence_length):
    tokens = tokenizer(text)
    tokens = [vocab[token] for token in tokens]

    num_pads = sequence_length - len(tokens)
    tokens = tokens + [vocab["<pad>"]] * num_pads

    return torch.tensor(tokens, dtype=torch.long)

# Vectorize the samples
sequence_length = 5
vectorized_sample1 = vectorize(sample1,
                               vocab,
                               sequence_length)
vectorized_sample2 = vectorize(sample2,
                               vocab,
                               sequence_length)

print("Vectorized Sample 1:", vectorized_sample1)
print("Vectorized Sample 2:", vectorized_sample2)
```

```
Vectorized Sample 1: tensor([0, 4, 7, 2, 1])
Vectorized Sample 2: tensor([2, 6, 3, 5, 0])
```

Embedding

- Example corpus
  - sample1: ‘We are learning AI’
  - sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

(2) Transform text into features

We are learning AI					AI is a CS topic				
we	are	learning	ai		ai	is	a	cs	topic
0	4	7	2	1	2	6	3	5	0

Standardize

Vectorization

```python
def vectorize(text, vocab, seq_len):
    tokens = tokenizer(text)
    tokens = [vocab[token] for token in tokens]

    num_pads = sequence_length - len(tokens)
    tokens = tokens[:sequence_length]
            + [vocab["<pad>"]]*num_pads

    return torch.tensor(tokens, dtype=torch.long)

# Vectorize the samples
sequence_length = 5
vectorized_sample1 = vectorize(sample1, vocab,
                               sequence_length)
vectorized_sample2 = vectorize(sample2, vocab,
                               sequence_length)


print("Vectorized Sample 1:", vectorized_sample1)
print("Vectorized Sample 2:", vectorized_sample2)
```

```
Vectorized Sample 1: tensor([0, 4, 7, 2, 1])
Vectorized Sample 2: tensor([2, 6, 3, 5, 0])
```

```python
sample3 = 'AI topic in CS is difficult'
vectorized_sample3 = vectorize(sample3, vocab,
                               sequence_length)
print(vectorized_sample3)
```

```
tensor([2, 0, 0, 5, 6])
```

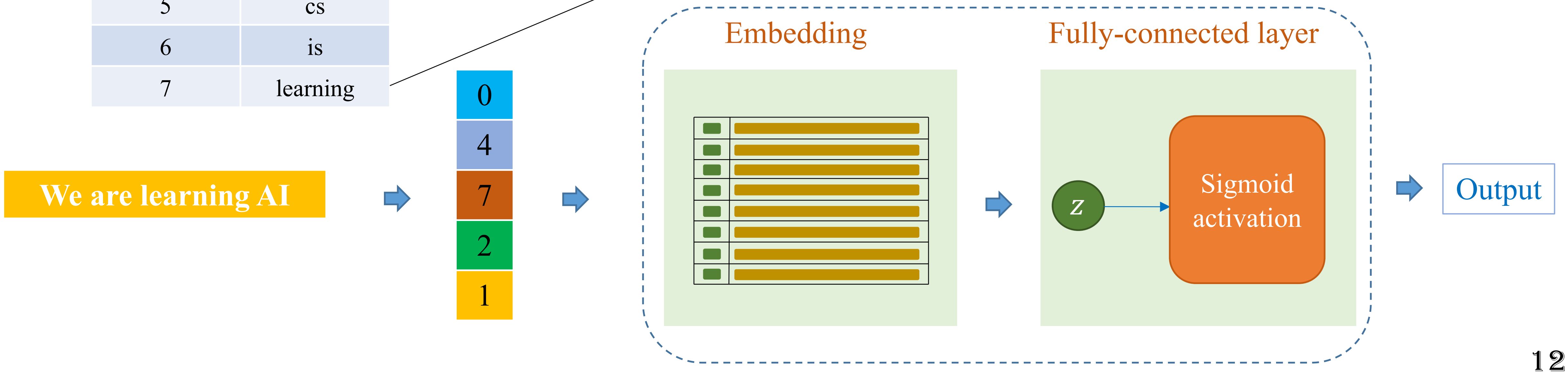
Embedding Layer

```
vocab_size = 8
embed_dim = 4
embedding = nn.Embedding(vocab_size,
                         embed_dim)
```

```
Parameter containing:
tensor([[-0.1882,  0.5530,  1.6267,  0.7013],
        [ 1.7840, -0.8278, -0.2701,  1.3586],
        [ 1.0281, -1.9094,  0.3182,  0.4211],
        [-1.3083, -0.0987,  0.7647, -0.3680],
        [ 0.2293,  1.3255,  0.1318,  2.0501],
        [ 0.4058, -0.6624, -0.8745,  0.7203],
        [ 0.5582,  0.0786, -0.6817,  0.6902],
        [ 0.4309, -1.3067, -0.8823,  1.5977]],
```

(3) Embedding layer

index	word
0	[UNK]
1	[pad]
2	ai
3	a
4	are
5	cs
6	is
7	learning



Embedding Layer

(3) Embedding layer

index	word
0	[UNK]
1	[pad]
2	ai
3	a
4	are
5	cs
6	is
7	learning

We are learning AI →

0
4
7
2
1

changed

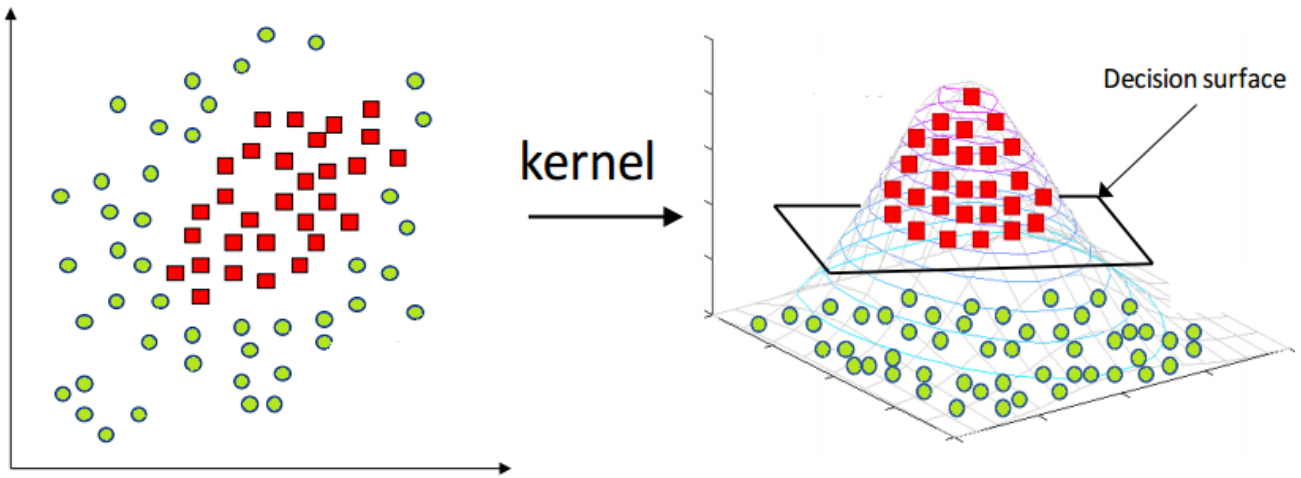
```
Parameter containing:
tensor([[-0.1882,  0.5530,  1.6267,  0.7013],
        [ 1.7840, -0.8278, -0.2701,  1.3586],
        [ 1.0281, -1.9094,  0.3182,  0.4211],
        [-1.3083, -0.0987,  0.7647, -0.3680],
        [ 0.2293,  1.3255,  0.1318,  2.0501],
        [ 0.4058, -0.6624, -0.8745,  0.7203],
        [ 0.5582,  0.0786, -0.6817,  0.6902],
        [ 0.4309, -1.3067, -0.8823,  1.5977]],
```

```
Parameter containing:
tensor([[-0.1872,  0.5540,  1.6277,  0.7023],
        [ 1.7830, -0.8268, -0.2711,  1.3576],
        [ 1.0291, -1.9084,  0.3192,  0.4201],
        [-1.3083, -0.0987,  0.7647, -0.3680],
        [ 0.2303,  1.3245,  0.1308,  2.0511],
        [ 0.4058, -0.6624, -0.8745,  0.7203],
        [ 0.5582,  0.0786, -0.6817,  0.6902],
        [ 0.4299, -1.3077, -0.8833,  1.5967]],
```

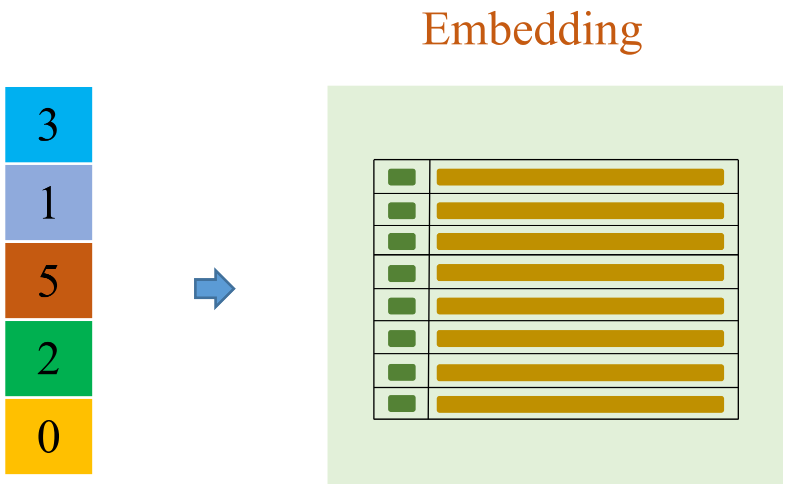
Word Embedding

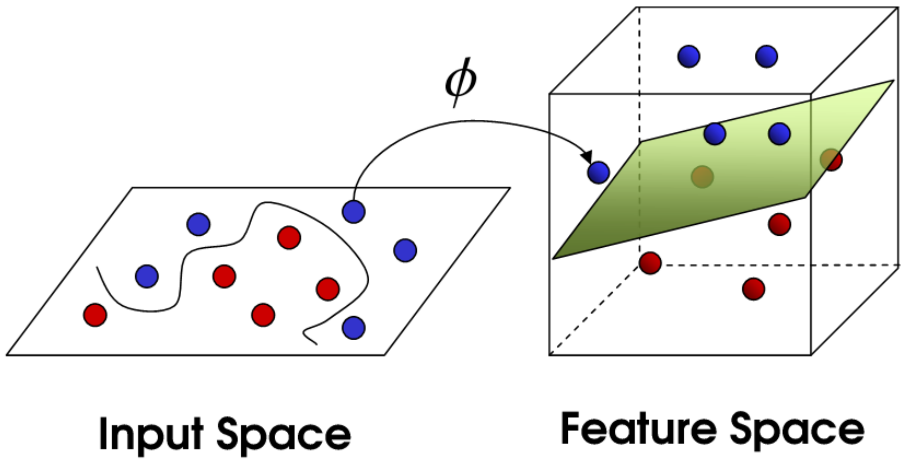
❖ **Why?**

1) Continuous values
2) Higher dimensions



https://codatalicious.medium.com/kernels-ee967067aa9





https://towardsdatascience.com/the-kernel-trick-c98cdbcaeb3f

Outline

SECTION 1

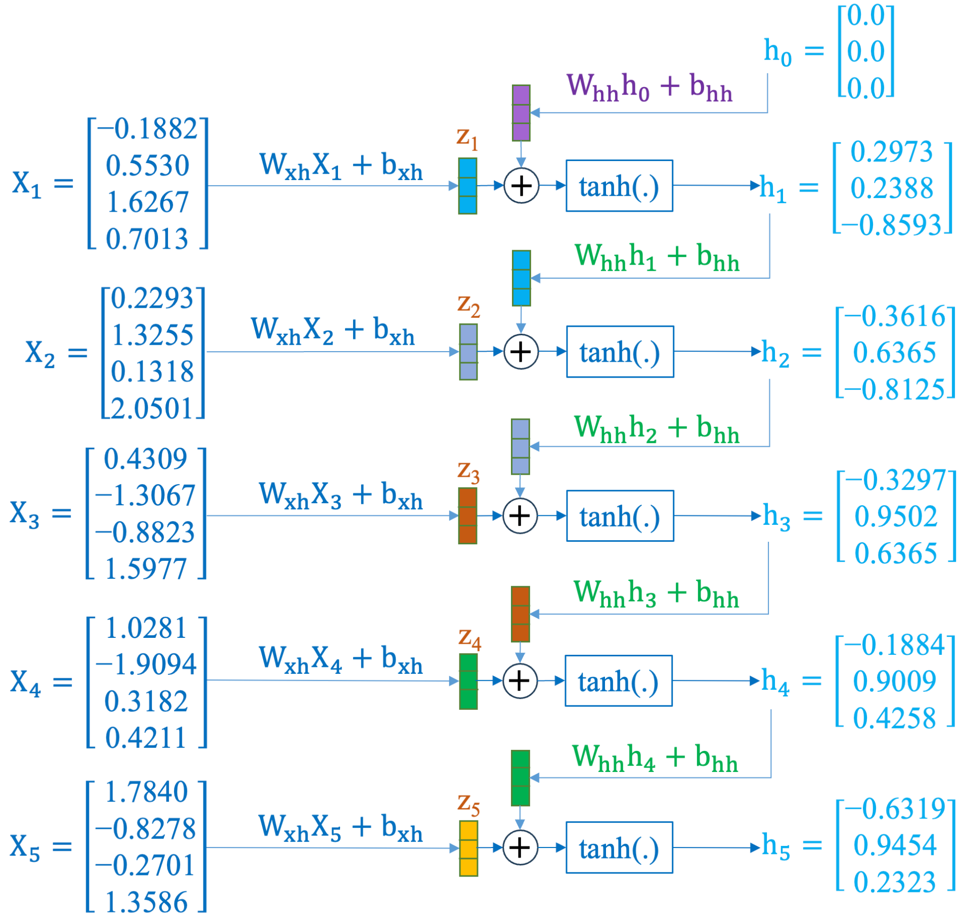
Text Classification

SECTION 2

RNN Construction

SECTION 3

RNNs for Time-series



Revisit input x

Convert from text to numbers

index	word
0	[UNK]
1	[pad]
2	ai
3	a
4	are
5	cs
6	is
7	learning

A sample X

We are learning AI → [0, 4, 7, 2, 1]

```
Parameter containing:
tensor([[-0.1882,  0.5530,  1.6267,  0.7013],
        [ 1.7840, -0.8278, -0.2701,  1.3586],
        [ 1.0281, -1.9094,  0.3182,  0.4211],
        [-1.3083, -0.0987,  0.7647, -0.3680],
        [ 0.2293,  1.3255,  0.1318,  2.0501],
        [ 0.4058, -0.6624, -0.8745,  0.7203],
        [ 0.5582,  0.0786, -0.6817,  0.6902],
        [ 0.4309, -1.3067, -0.8823,  1.5977]],
```

A sample X

We	0	X_1: [-0.1882, 0.5530, 1.6267, 0.7013]
are	4	X_2: [0.2293, 1.3255, 0.1318, 2.0501]
learning	7	X_3: [0.4309, -1.3067, -0.8823, 1.5977]
AI	2	X_4: [1.0281, -1.9094, 0.3182, 0.4211]
<pad>	1	X_5: [1.7840, -0.8278, -0.2701, 1.3586]

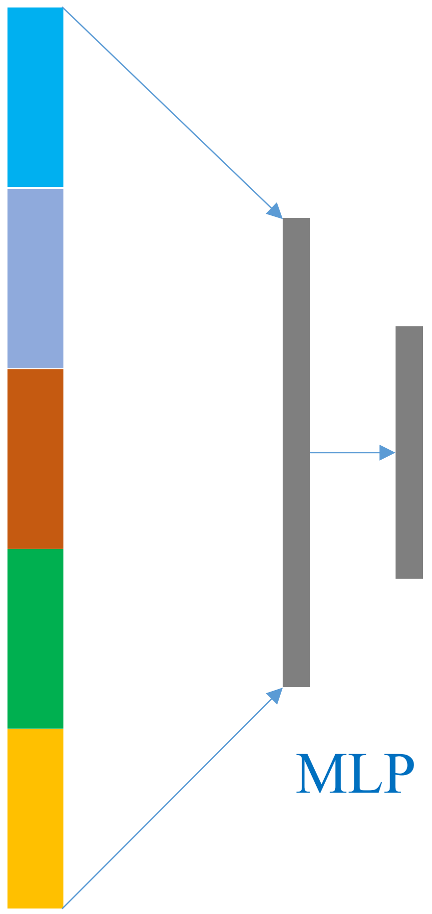
How to feed X to a network?
(Get ideas from how MLP and CNN work)

How to deal with this input?

- ❖ **Simplest idea: Based on MLP**
- ❖ **Concatenate all the features**

We	0	X_1: [-0.1882, 0.5530, 1.6267, 0.7013]
are	4	X_2: [0.2293, 1.3255, 0.1318, 2.0501]
learning	7	X_3: [0.4309, -1.3067, -0.8823, 1.5977]
AI	2	X_4: [1.0281, -1.9094, 0.3182, 0.4211]
<pad>	1	X_5: [1.7840, -0.8278, -0.2701, 1.3586]

A sample X



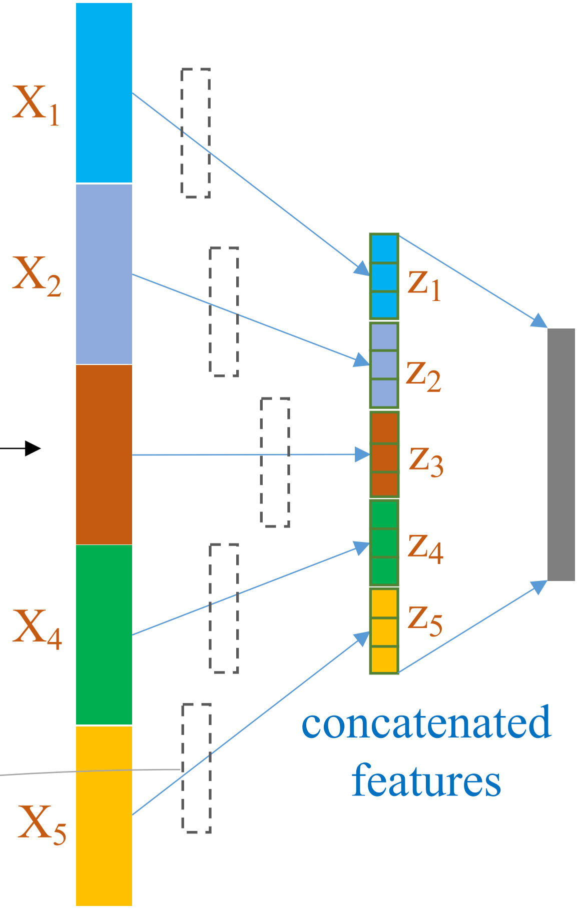
concatenated features

Pros and cons?

❖ Based on CNN: Shared weights

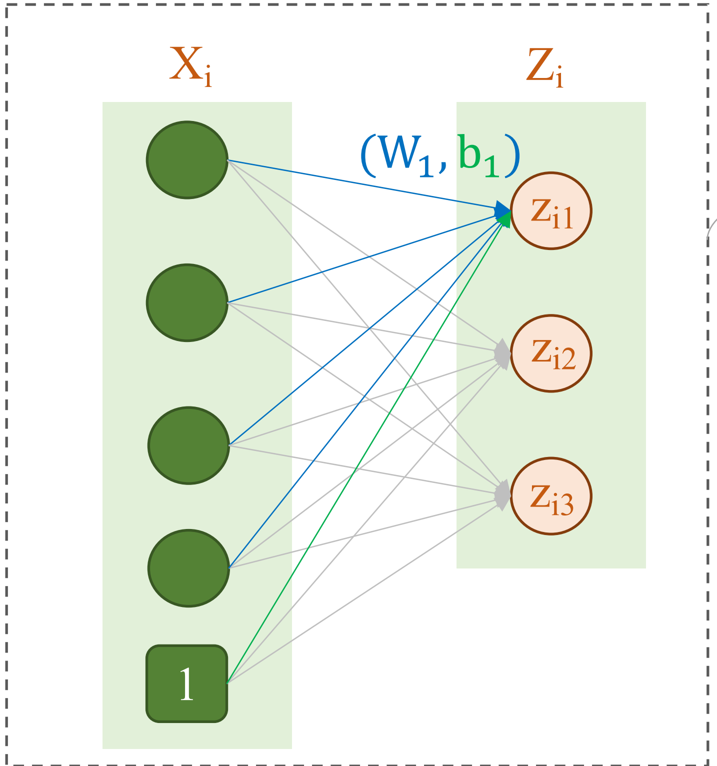
We	0	X_1: [-0.1882, 0.5530, 1.6267, 0.7013]
are	4	X_2: [0.2293, 1.3255, 0.1318, 2.0501]
learning	7	X_3: [0.4309, -1.3067, -0.8823, 1.5977]
AI	2	X_4: [1.0281, -1.9094, 0.3182, 0.4211]
<pad>	1	X_5: [1.7840, -0.8278, -0.2701, 1.3586]

A sample X



$W =$

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad z_i = \begin{bmatrix} z_{i1} \\ z_{i2} \\ z_{i3} \end{bmatrix}$$



$$z_{i1} = W_1{}^T X_i + b_1$$
$$z_{i2} = W_2{}^T X_i + b_2$$
$$z_{i3} = W_3{}^T X_i + b_3$$

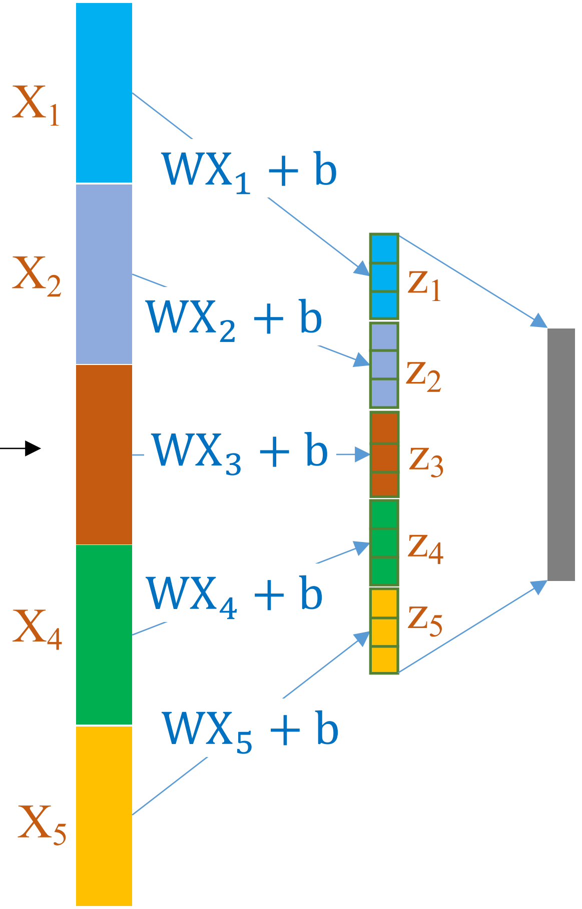
Fully-connected layer
Shared weights

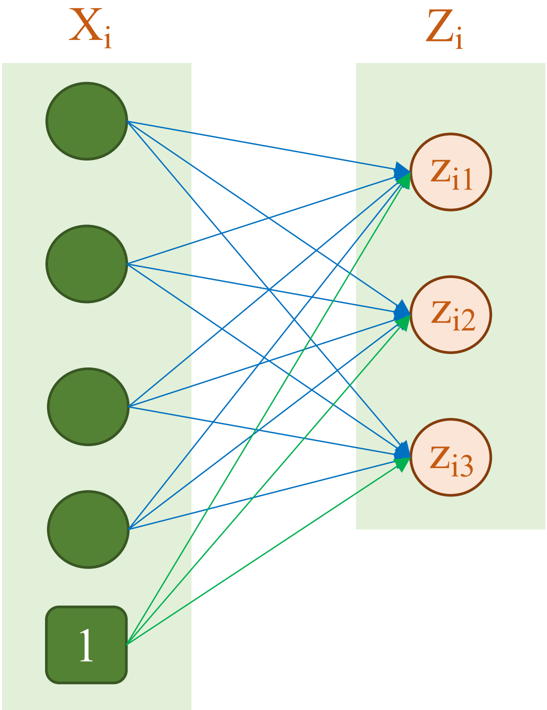
➡ $z_i = WX_i + b$

❖ Based on CNN: Shared weights

A sample X

We	0	X_1: [-0.1882, 0.5530, 1.6267, 0.7013]
are	4	X_2: [0.2293, 1.3255, 0.1318, 2.0501]
learning	7	X_3: [0.4309, -1.3067, -0.8823, 1.5977]
AI	2	X_4: [1.0281, -1.9094, 0.3182, 0.4211]
<pad>	1	X_5: [1.7840, -0.8278, -0.2701, 1.3586]





$$z_{i1} = W_1{}^T X_i + b_1$$
$$z_{i2} = W_2{}^T X_i + b_2$$
$$z_{i3} = W_3{}^T X_i + b_3$$

$$\Rightarrow z_i = WX_i + b$$

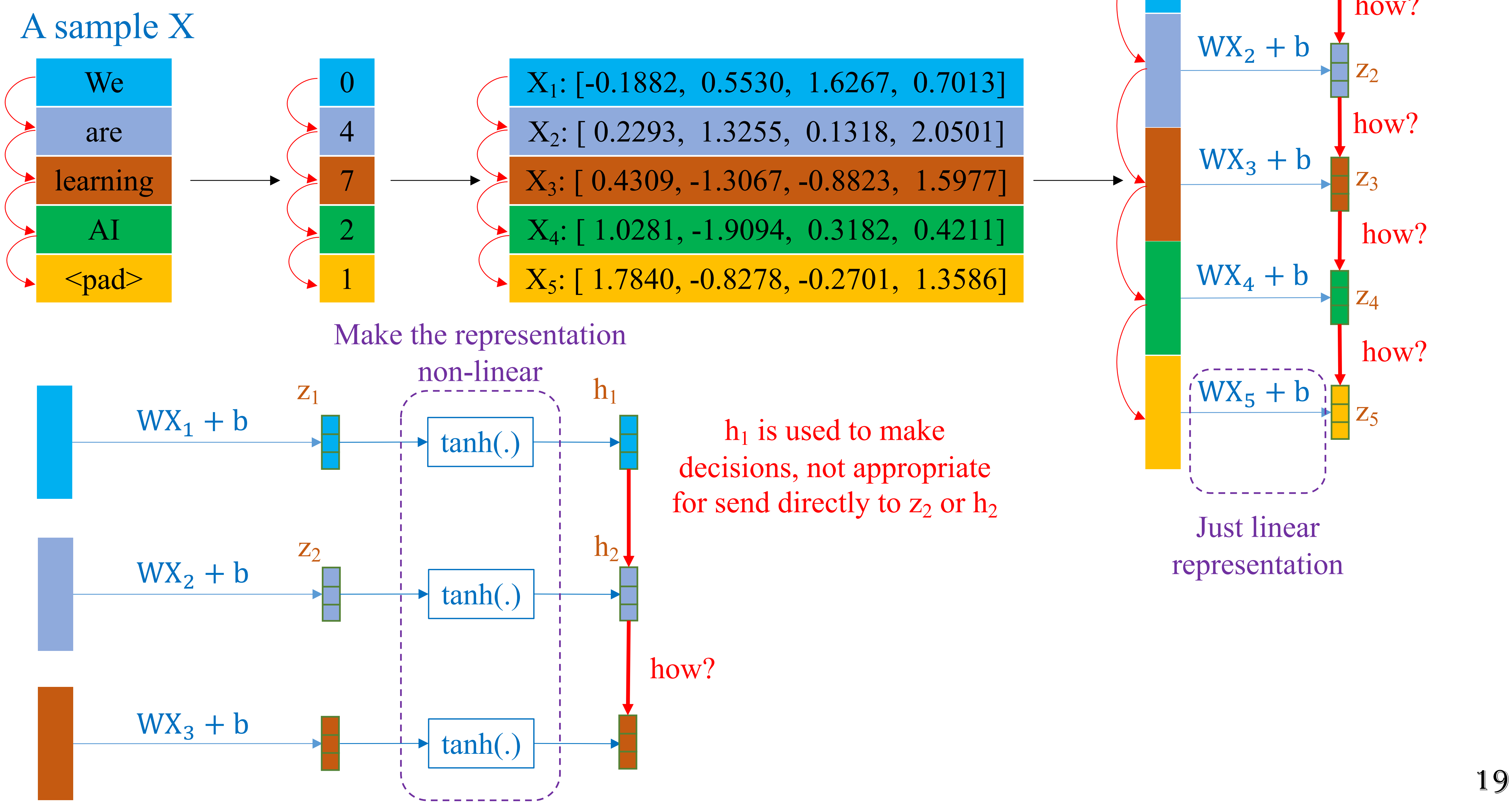
$W =$ 

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad z_i = \begin{bmatrix} z_{i1} \\ z_{i2} \\ z_{i3} \end{bmatrix}$$

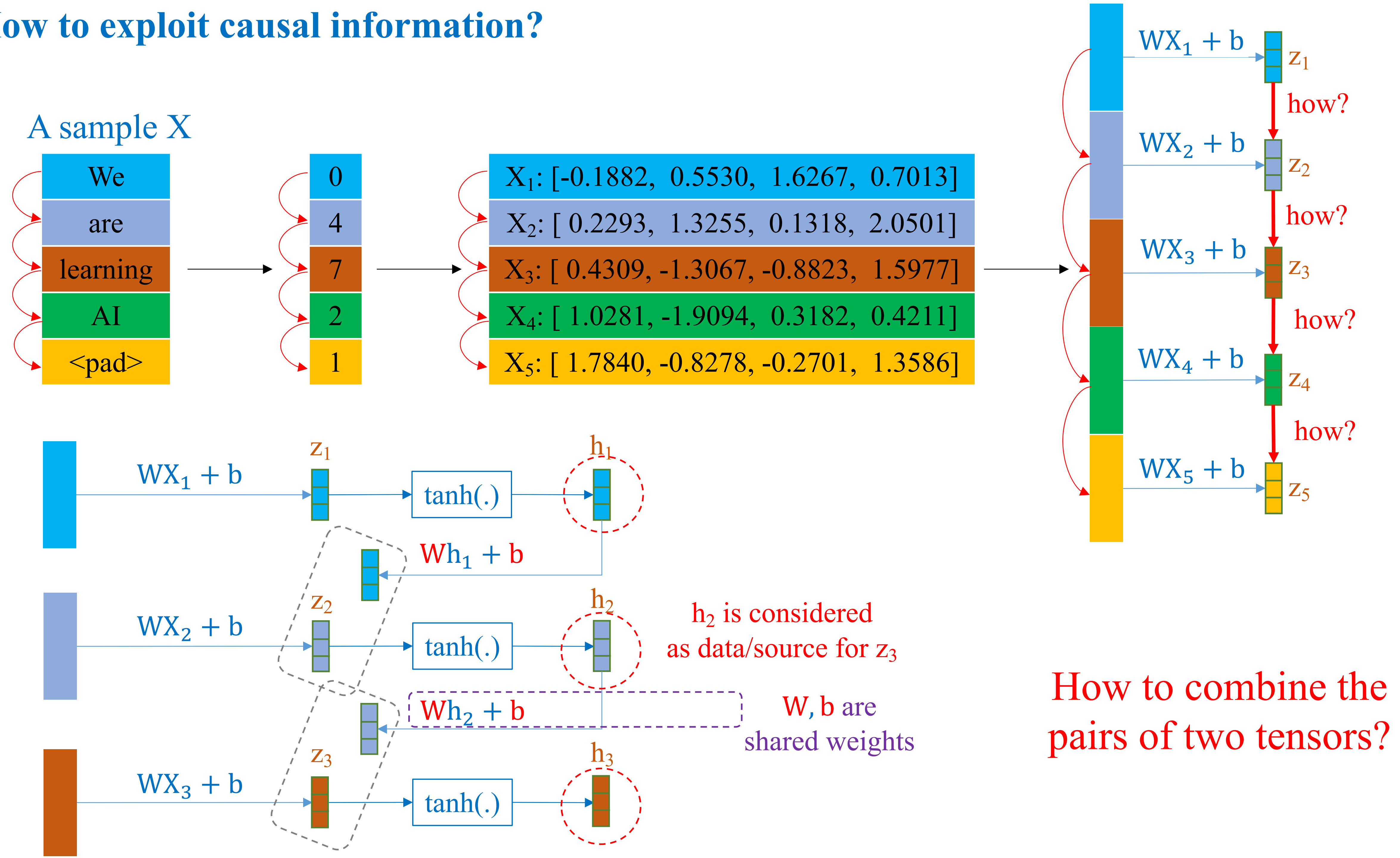
$$z_1 = WX_1 + b$$
$$z_2 = WX_2 + b$$
$$z_3 = WX_3 + b$$
$$z_4 = WX_4 + b$$
$$z_5 = WX_5 + b$$

Any thing missing?

❖ How to exploit causal information?



❖ How to exploit causal information?



❖ How to exploit causal information?

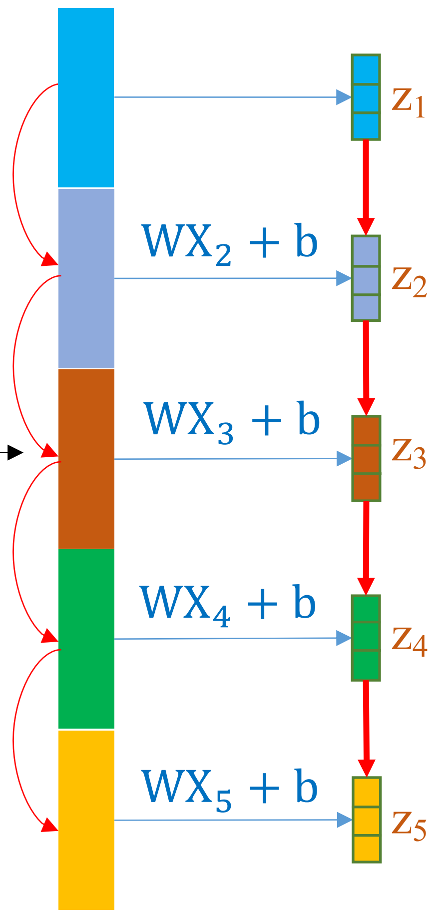
X_1: [-0.1882, 0.5530, 1.6267, 0.7013]

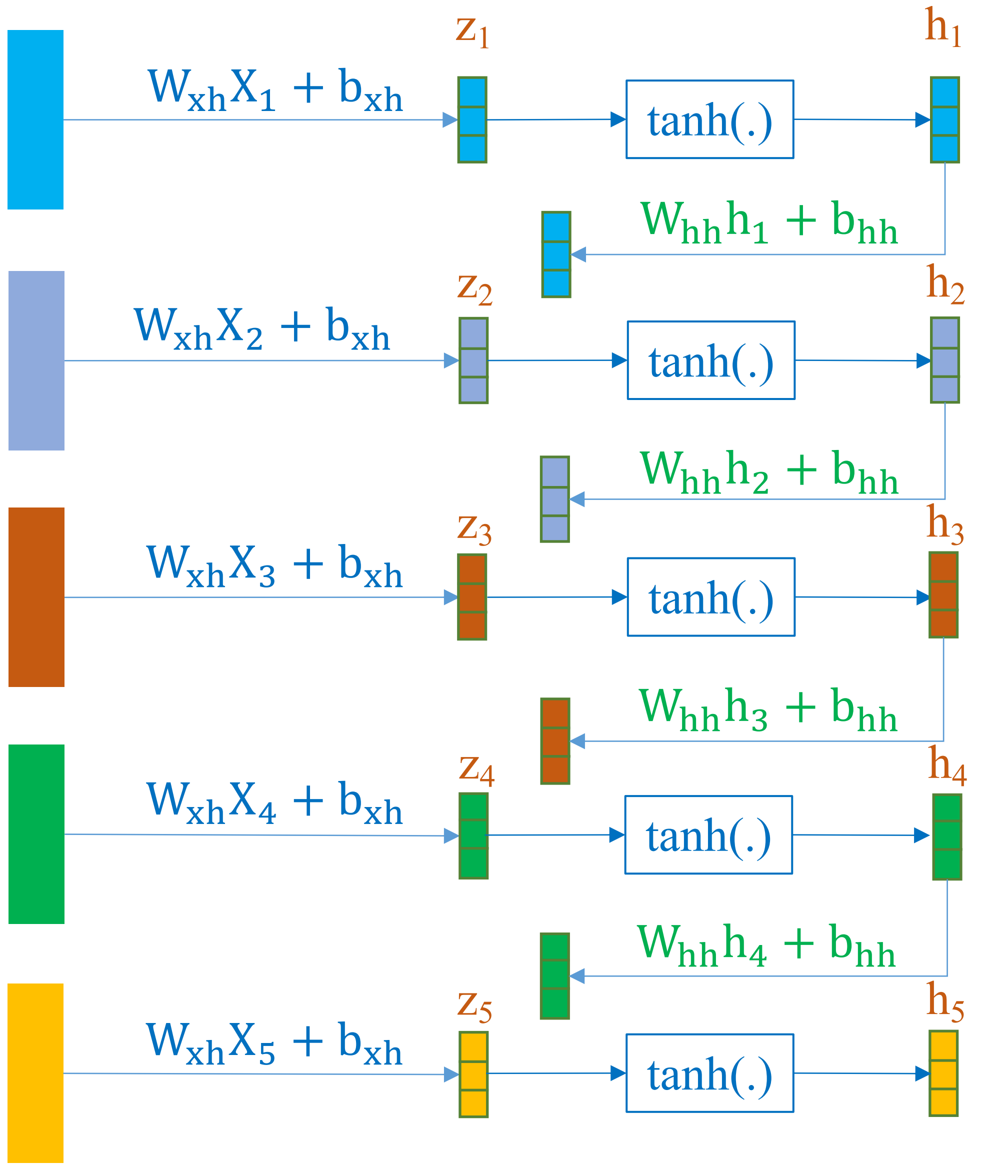
X_2: [0.2293, 1.3255, 0.1318, 2.0501]

X_3: [0.4309, -1.3067, -0.8823, 1.5977]

X_4: [1.0281, -1.9094, 0.3182, 0.4211]

X_5: [1.7840, -0.8278, -0.2701, 1.3586]

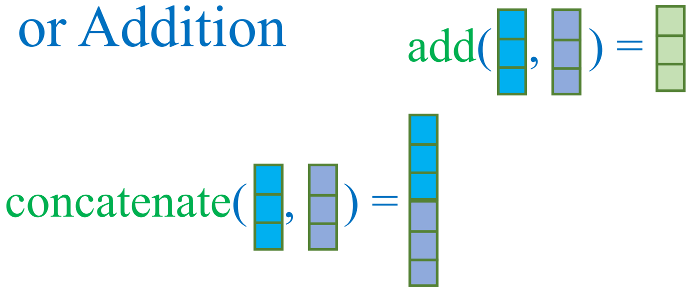




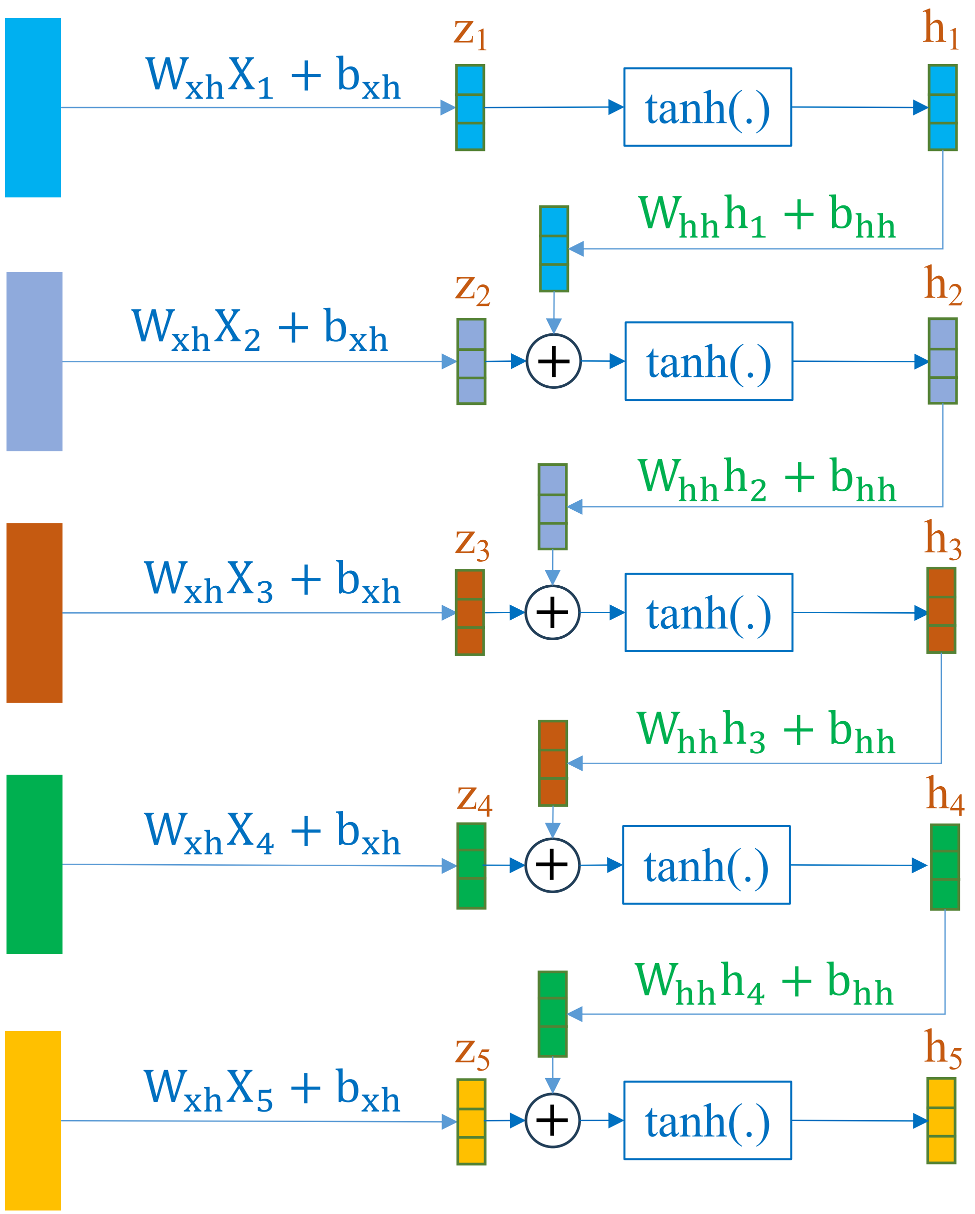
How to combine two tensors?

Idea from skip-connection

- or Concatenation
- or Addition



❖ How to exploit causal information?



$$h_1 = \tanh(W_{xh}X_1 + b_{xh})$$

$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh})$$

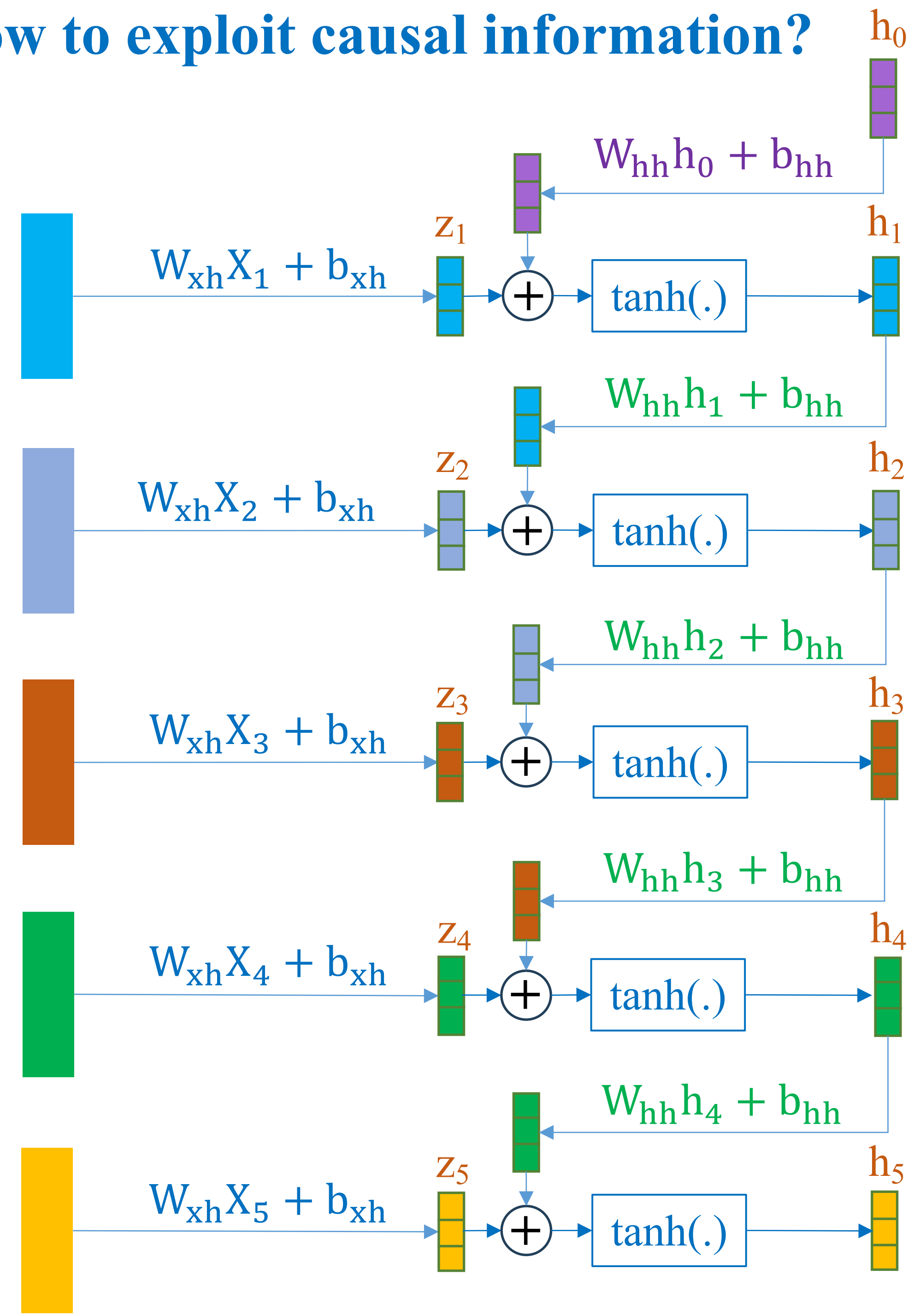
$$h_3 = \tanh(W_{xh}X_3 + b_{xh} + W_{hh}h_2 + b_{hh})$$

$$h_4 = \tanh(W_{xh}X_4 + b_{xh} + W_{hh}h_3 + b_{hh})$$

$$h_5 = \tanh(W_{xh}X_5 + b_{xh} + W_{hh}h_4 + b_{hh})$$

h_1 formula is different from the others. How to solve this issue?

❖ How to exploit causal information?



$$h_0 = \mathbf{0} \qquad b_{hh} = \mathbf{0}$$

$$h_1 = \tanh(W_{xh}X_1 + b_{xh} + W_{hh}h_0 + b_{hh})$$

$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh})$$

$$h_3 = \tanh(W_{xh}X_3 + b_{xh} + W_{hh}h_2 + b_{hh})$$

$$h_4 = \tanh(W_{xh}X_4 + b_{xh} + W_{hh}h_3 + b_{hh})$$

$$h_5 = \tanh(W_{xh}X_5 + b_{xh} + W_{hh}h_4 + b_{hh})$$

➡ $h_t = \tanh\left(W_{xh}X_t + b_{xh} + W_{hh}h_{(t-1)} + b_{hh}\right)$

➡ RNN

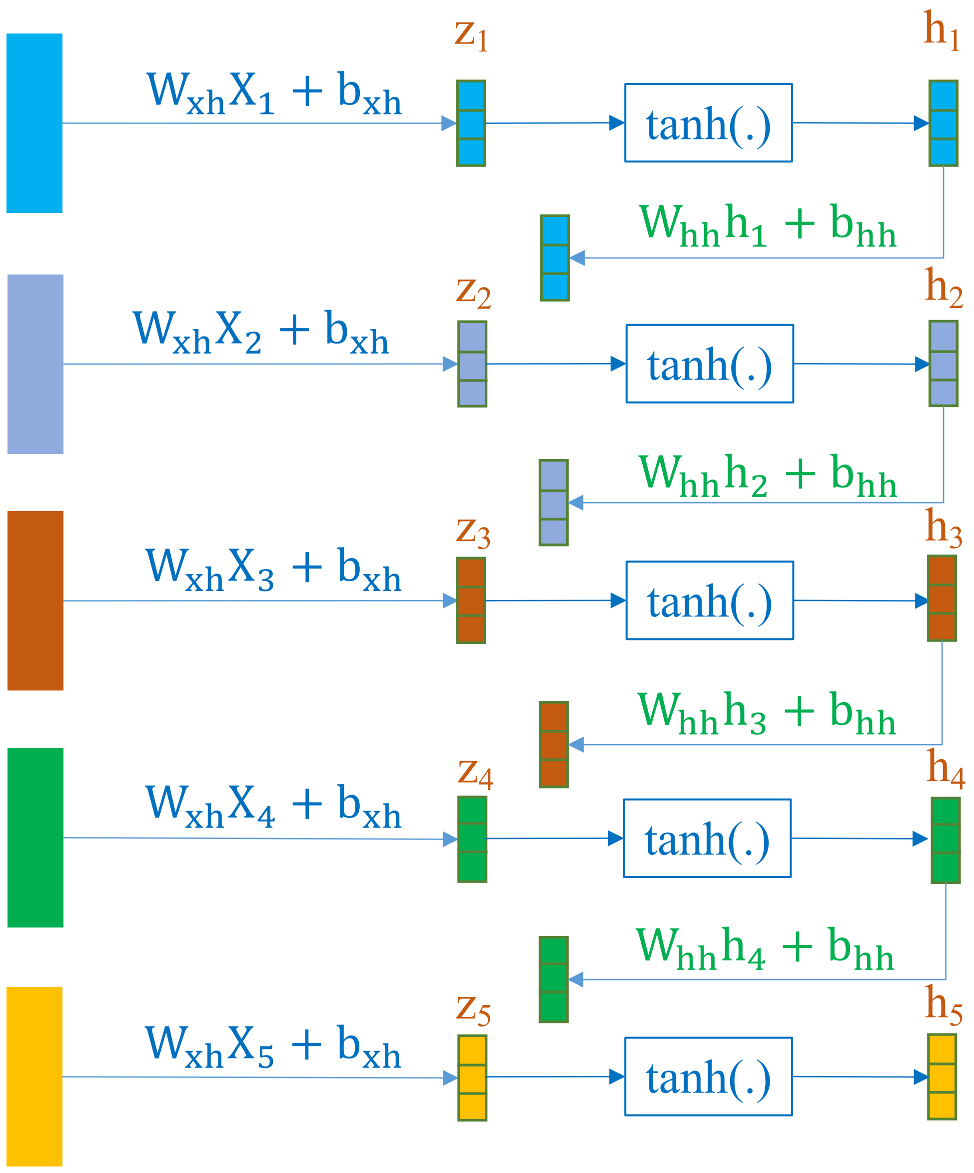
Example

X_1: [-0.1882, 0.5530, 1.6267, 0.7013]
X_2: [0.2293, 1.3255, 0.1318, 2.0501]
X_3: [0.4309, -1.3067, -0.8823, 1.5977]
X_4: [1.0281, -1.9094, 0.3182, 0.4211]
X_5: [1.7840, -0.8278, -0.2701, 1.3586]

$$X_1 = \begin{bmatrix} -0.1882 \\ 0.5530 \\ 1.6267 \\ 0.7013 \end{bmatrix} \qquad X_2 = \begin{bmatrix} 0.2293 \\ 1.3255 \\ 0.1318 \\ 2.0501 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} 0.4309 \\ -1.3067 \\ -0.8823 \\ 1.5977 \end{bmatrix} \qquad X_4 = \begin{bmatrix} 1.0281 \\ -1.9094 \\ 0.3182 \\ 0.4211 \end{bmatrix}$$

$$X_5 = \begin{bmatrix} 1.7840 \\ -0.8278 \\ -0.2701 \\ 1.3586 \end{bmatrix}$$



Example

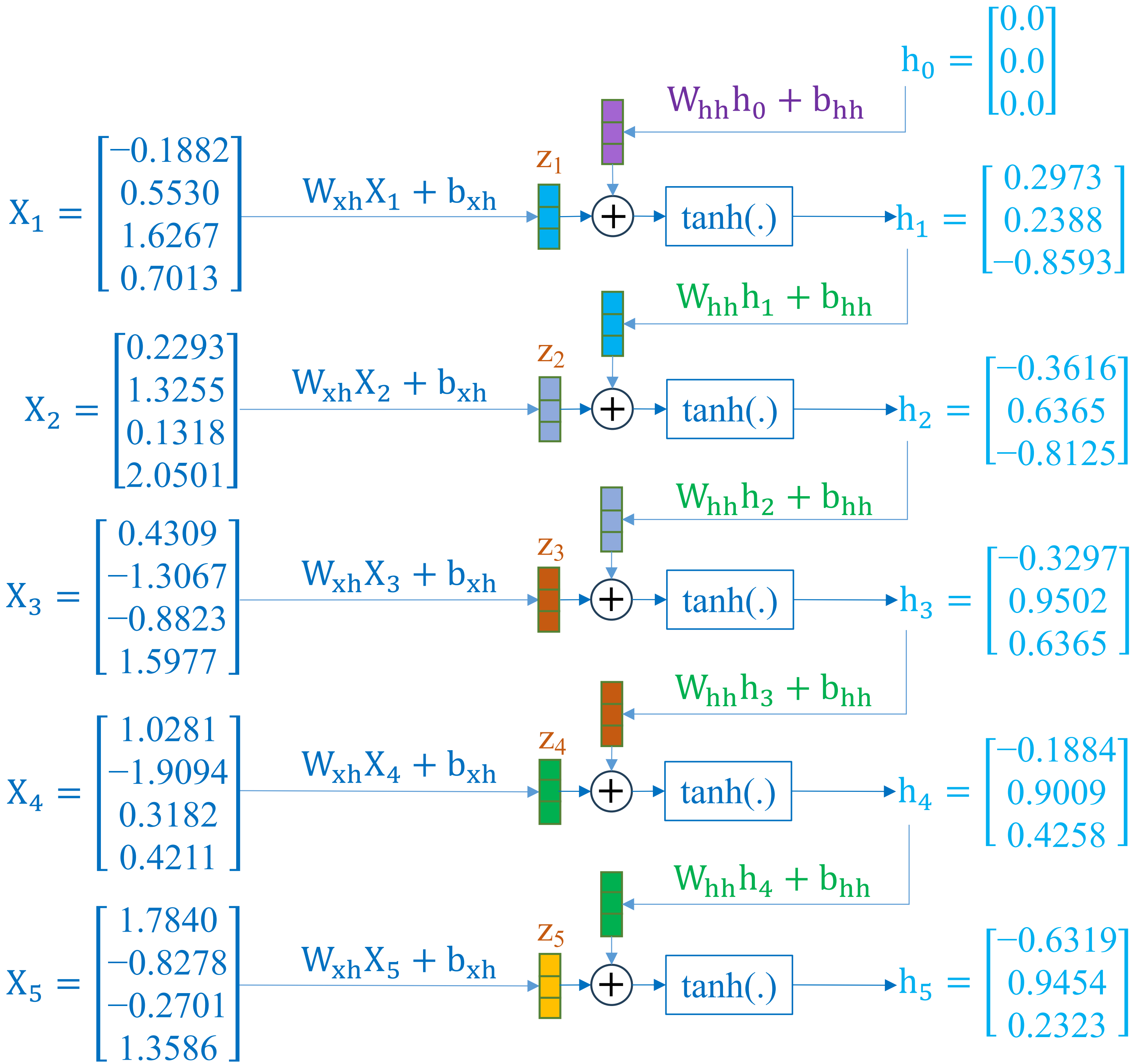
$$W_{xh} = \begin{bmatrix} -0.4174 & 0.1953 & -0.0365 & -0.4025 \\ 0.4722 & -0.4085 & -0.0236 & 0.3763 \\ 0.0550 & -0.4921 & -0.4307 & 0.0855 \end{bmatrix}$$

$$b_{xh} = \begin{bmatrix} 0.4481 \\ 0.5537 \\ -0.5006 \end{bmatrix}$$

$$W_{hh} = \begin{bmatrix} -0.5511 & 0.1260 & 0.0463 \\ -0.2291 & -0.2084 & -0.2352 \\ -0.4341 & -0.2682 & 0.0612 \end{bmatrix}$$

$$b_{hh} = \begin{bmatrix} 0.0135 \\ -0.2209 \\ 0.1330 \end{bmatrix}$$

2.i1_RNN_Layer.ipynb



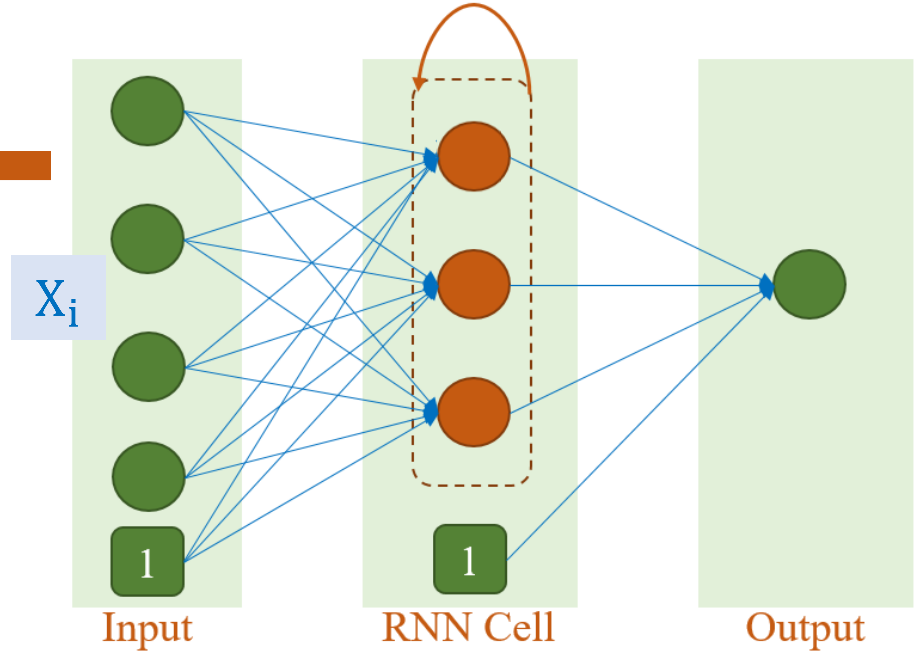
Text Deep Models

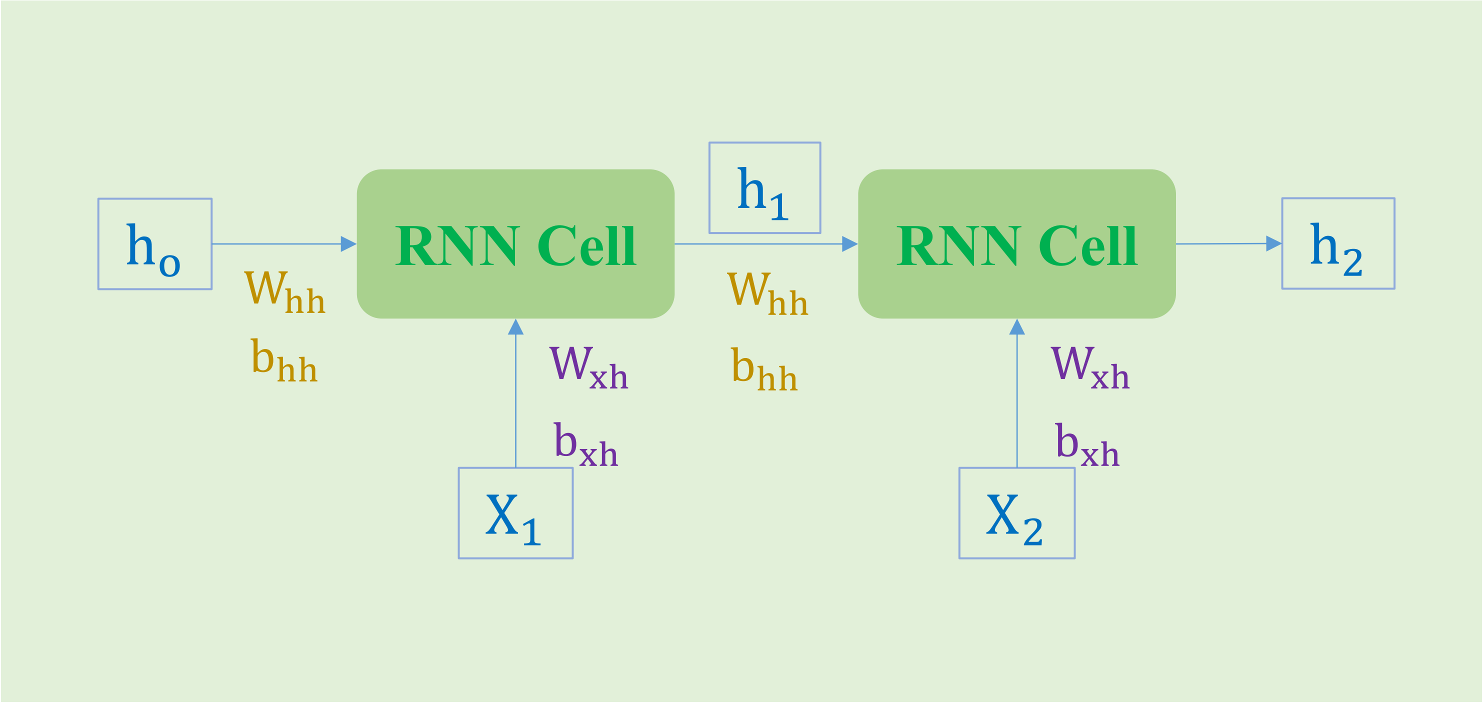
❖ **Recurrent Neural Networks (RNNs)**

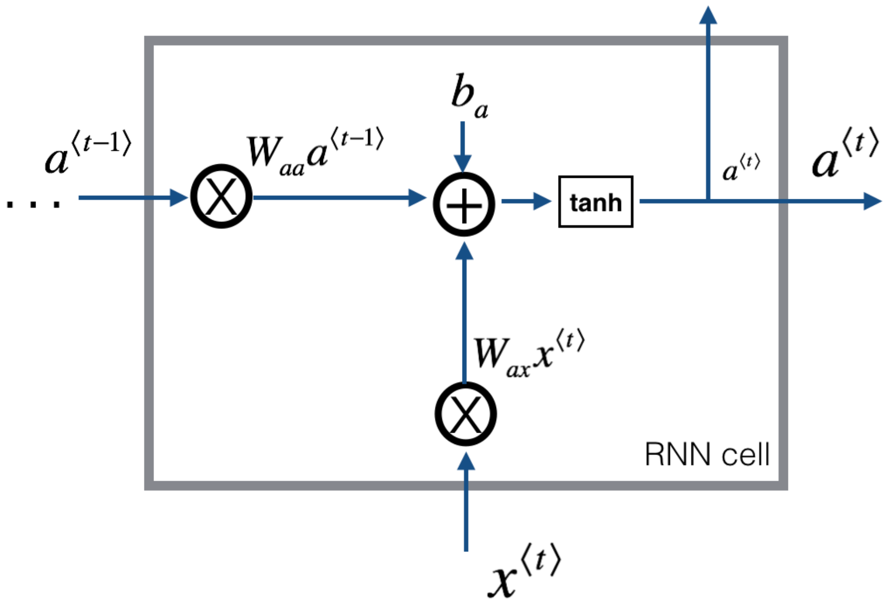
❖ **Classification and time-step=2**

$$h_1 = \tanh(W_{xh}X_1 + b_{xh} + W_{hh}h_0 + b_{hh})$$

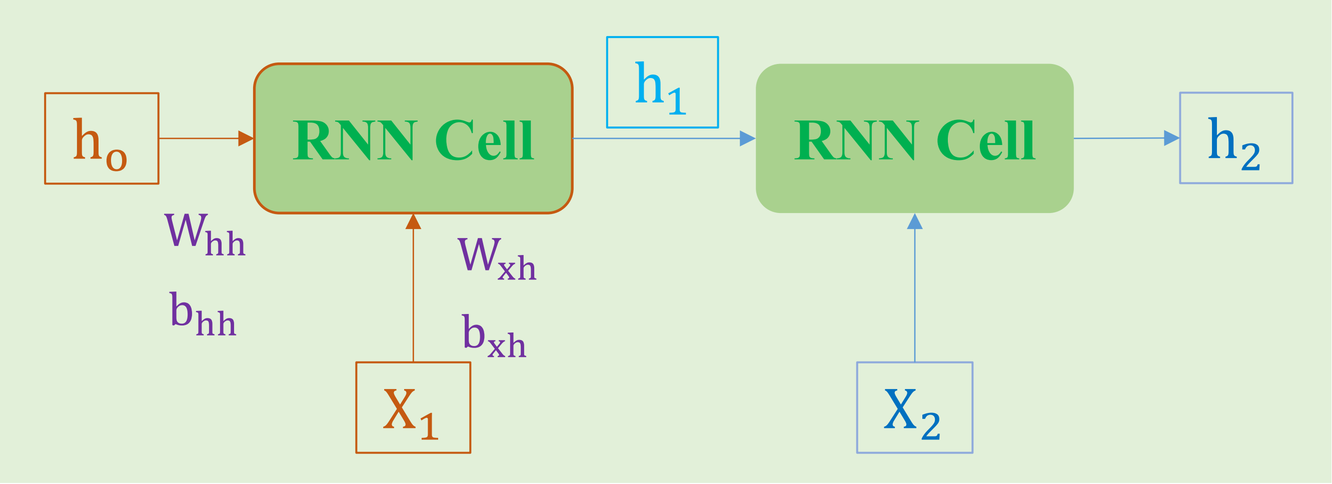
$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh})$$







https://datascience-enthusiast.com/DL/Building_a_Recurrent_Neural_Network-Step_by_Step_v1.html



$$\mathbf{W_{hh}}=\begin{bmatrix} 0.9 & -0.1 & 0.1 \\ -0.1 & -0.9 & 0.3 \\ 0.1 & -0.3 & -0.9 \end{bmatrix} \qquad \mathbf{b_{hh}}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

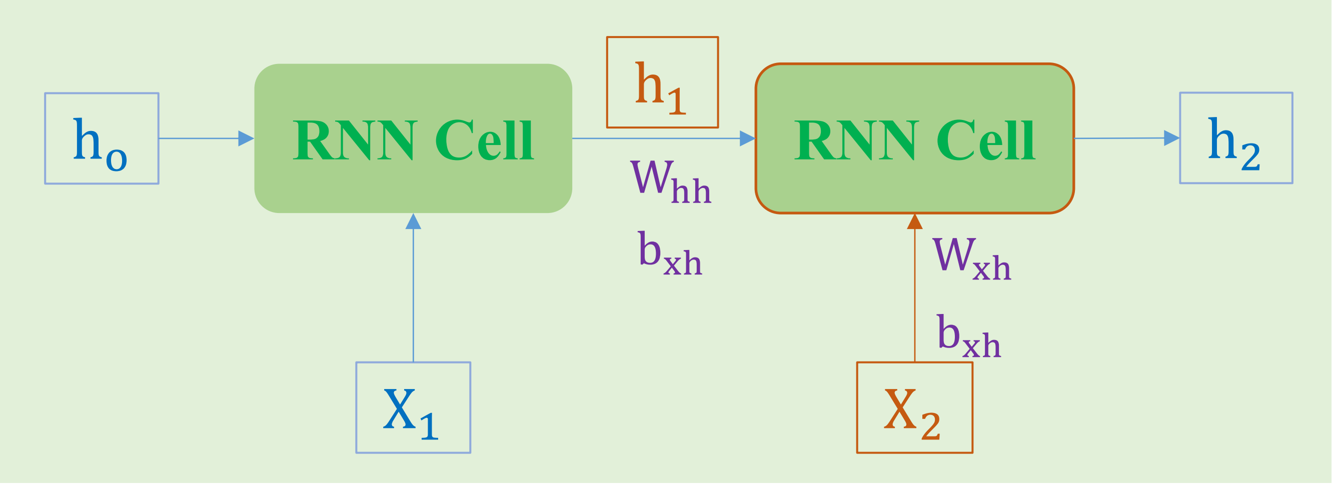
$$\mathbf{W_{xh}}=\begin{bmatrix} -0.2 & -0.4 & 0.3 & -0.4 \\ -0.6 & -0.8 & 0.5 & -0.3 \\ 0.1 & -0.1 & 0.6 & 0.1 \end{bmatrix} \qquad \mathbf{b_{xh}}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$\mathbf{X}=\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}=\begin{bmatrix} 1. & 3. & 2. & 1. \\ 0. & 4. & 1. & 2. \end{bmatrix} \qquad \mathbf{h_0}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$h_1 = \tanh(W_{xh}X_1 + b_{xh} + W_{hh}h_0 + b_{hh})$$

$$= \tanh\left(\begin{bmatrix} -0.2 & -0.4 & 0.3 & -0.4 \\ -0.6 & -0.8 & 0.5 & -0.3 \\ 0.1 & -0.1 & 0.6 & 0.1 \end{bmatrix}\begin{bmatrix} 1.0 \\ 3.0 \\ 2.0 \\ 1.0 \end{bmatrix} + \begin{bmatrix} 0.9 & -0.1 & 0.1 \\ -0.1 & -0.9 & 0.3 \\ 0.1 & -0.3 & -0.9 \end{bmatrix}\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} + \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}\right)$$

$$= \tanh\left(\begin{bmatrix} -1.2 \\ -2.3 \\ 1.1 \end{bmatrix}\right) = \begin{bmatrix} -0.833 \\ -0.98 \\ 0.8 \end{bmatrix}$$



$$\mathbf{W_{hh}}=\begin{bmatrix} 0.9 & -0.1 & 0.1 \\ -0.1 & -0.9 & 0.3 \\ 0.1 & -0.3 & -0.9 \end{bmatrix} \qquad \mathbf{b_{hh}}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$\mathbf{W_{xh}}=\begin{bmatrix} -0.2 & -0.4 & 0.3 & -0.4 \\ -0.6 & -0.8 & 0.5 & -0.3 \\ 0.1 & -0.1 & 0.6 & 0.1 \end{bmatrix} \qquad \mathbf{b_{xh}}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$\mathbf{X}=\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}=\begin{bmatrix} 1. & 3. & 2. & 1. \\ 0. & 4. & 1. & 2. \end{bmatrix} \qquad \mathbf{h_0}=\begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$\boldsymbol{h}_1=\begin{bmatrix} -0.833 \\ -0.98 \\ 0.8 \end{bmatrix}$$

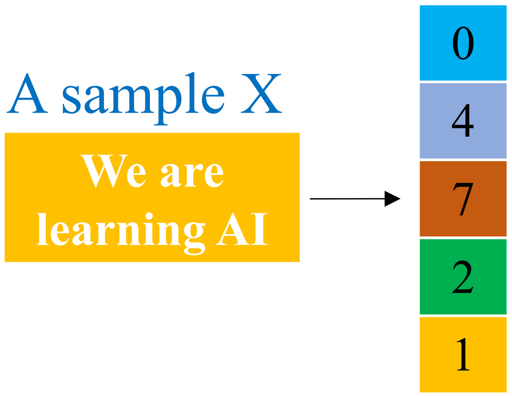
$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh})$$

$$= \tanh\left(\begin{bmatrix} -0.2 & -0.4 & 0.3 & -0.4 \\ -0.6 & -0.8 & 0.5 & -0.3 \\ 0.1 & -0.1 & 0.6 & 0.1 \end{bmatrix}\begin{bmatrix} 0.0 \\ 4.0 \\ 1.0 \\ 2.0 \end{bmatrix} + \begin{bmatrix} 0.9 & -0.1 & 0.1 \\ -0.1 & -0.9 & 0.3 \\ 0.1 & -0.3 & -0.9 \end{bmatrix}\begin{bmatrix} -0.833 \\ -0.98 \\ 0.8 \end{bmatrix} + \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}\right)$$

$$= \tanh\left(\begin{bmatrix} -2.67 \\ -2.09 \\ -0.109 \end{bmatrix}\right) = \begin{bmatrix} -0.99 \\ -0.97 \\ -0.109 \end{bmatrix}$$

RNN Models

❖ Implementation



(L, H_{out})

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \end{bmatrix}$$

Batch: (L, N, H_{out})

```
Parameter containing:
tensor([[-0.1882,  0.5530,  1.6267,  0.7013],
        [ 1.7840, -0.8278, -0.2701,  1.3586],
        [ 1.0281, -1.9094,  0.3182,  0.4211],
        [-1.3083, -0.0987,  0.7647, -0.3680],
        [ 0.2293,  1.3255,  0.1318,  2.0501],
        [ 0.4058, -0.6624, -0.8745,  0.7203],
        [ 0.5582,  0.0786, -0.6817,  0.6902],
        [ 0.4309, -1.3067, -0.8823,  1.5977]],
```

```
data = torch.tensor([0, 4, 7, 2, 1],
                    dtype=torch.long)
data_embedding = embedding(data)
print(data_embedding)
```

```
tensor([[-0.1882,  0.5530,  1.6267,  0.7013],
        [ 0.2293,  1.3255,  0.1318,  2.0501],
        [ 0.4309, -1.3067, -0.8823,  1.5977],
        [ 1.0281, -1.9094,  0.3182,  0.4211],
        [ 1.7840, -0.8278, -0.2701,  1.3586]],
```

```
embed_dim = 4
hidden_dim = 3
rnn = nn.RNN(embed_dim,
             hidden_dim,
             batch_first=True)

rnn_output, rnn_hidden = rnn(data_embedding)
rnn_output
```

```
tensor([[ 0.2973,  0.2388, -0.8593],
        [-0.3616,  0.6365, -0.8125],
        [-0.3297,  0.9502,  0.6365],
        [-0.1884,  0.9009,  0.4258],
        [-0.6319,  0.9454,  0.2323]], grad_fn=<
```

RNN Models

❖ Implementation

A sample X

We are learning AI → 0, 4, 7, 2, 1

(L, H_{out})

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \end{bmatrix}$$

Batch: (N, L, H_{out})

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \end{bmatrix} = \begin{bmatrix} [0.2973 & 0.2388 & -0.8593] \\ [-0.3616 & 0.6365 & -0.8125] \\ [-0.3297 & 0.9502 & 0.6365] \\ [-0.1884 & 0.9009 & 0.4258] \\ [-0.6319 & 0.9454 & 0.2323] \end{bmatrix}$$

```python
test_data = data_embedding.reshape(1, 5, 4)
print(test_data.shape)
```

```
torch.Size([1, 5, 4])
```

```python
rnn_output, rnn_hidden = rnn(test_data)

# (N, L, H_out)
print(rnn_output.shape)
print(rnn_output)
print(rnn_output[:, -1, :])
```

```
torch.Size([1, 5, 3])
tensor([[[ 0.2973,  0.2388, -0.8593],
         [-0.3616,  0.6365, -0.8125],
         [-0.3297,  0.9502,  0.6365],
         [-0.1884,  0.9009,  0.4258],
         [-0.6319,  0.9454,  0.2323]]], grad_fn=<T
tensor([[-0.6319,  0.9454,  0.2323]], grad_fn=<Sli
```

```python
# (num_layers, N, H_out)
print(rnn_hidden.shape)
print(rnn_hidden)
print(rnn_hidden[-1, :, :])
```

```
torch.Size([1, 1, 3])
tensor([[[-0.6319,  0.9454,  0.2323]]], grad_fn=<S
tensor([[-0.6319,  0.9454,  0.2323]], grad_fn=<Sli
```

Stack of RNNs

❖ Recurrent Neural Networks (RNNs)

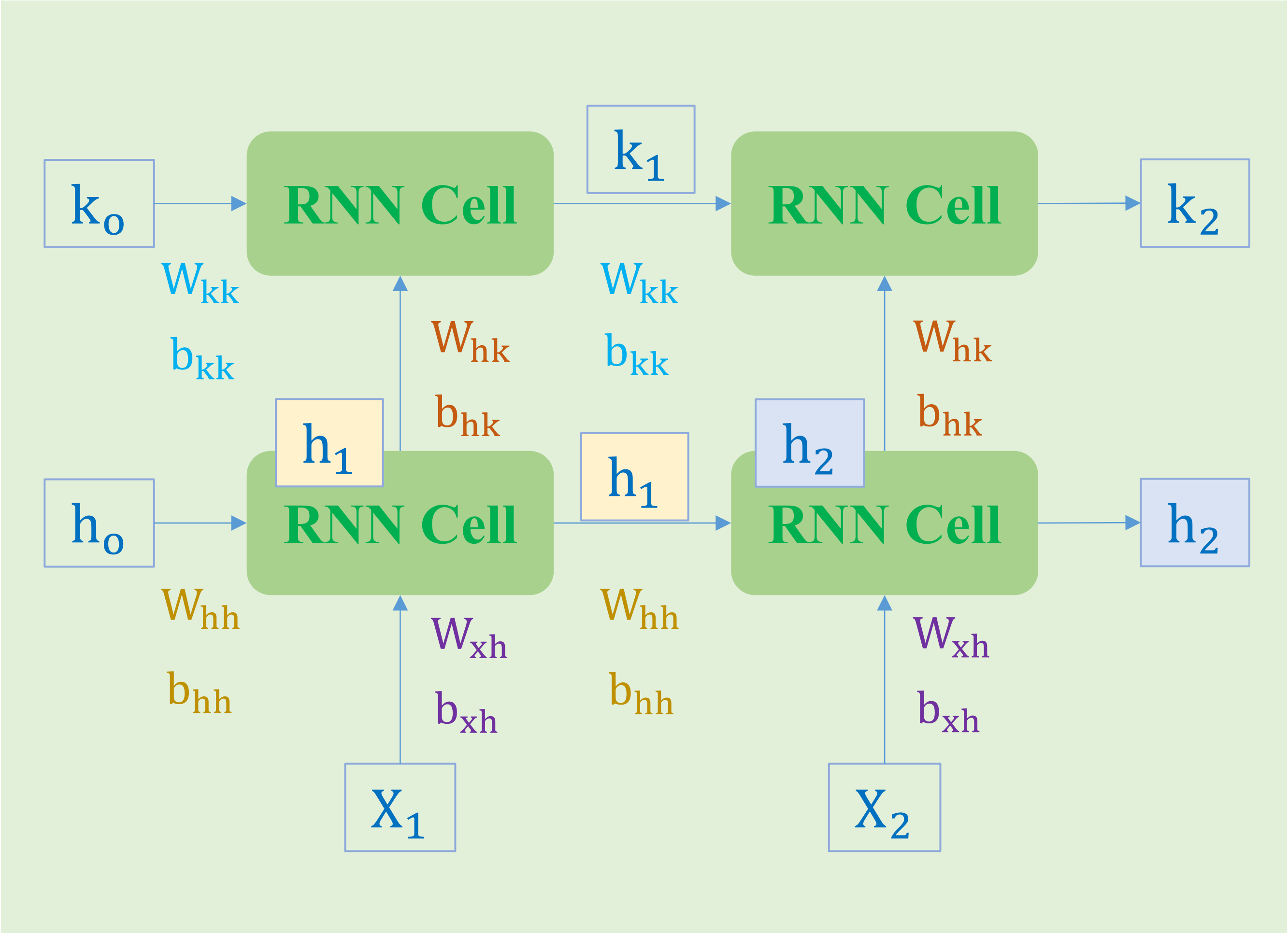
❖ Two layers

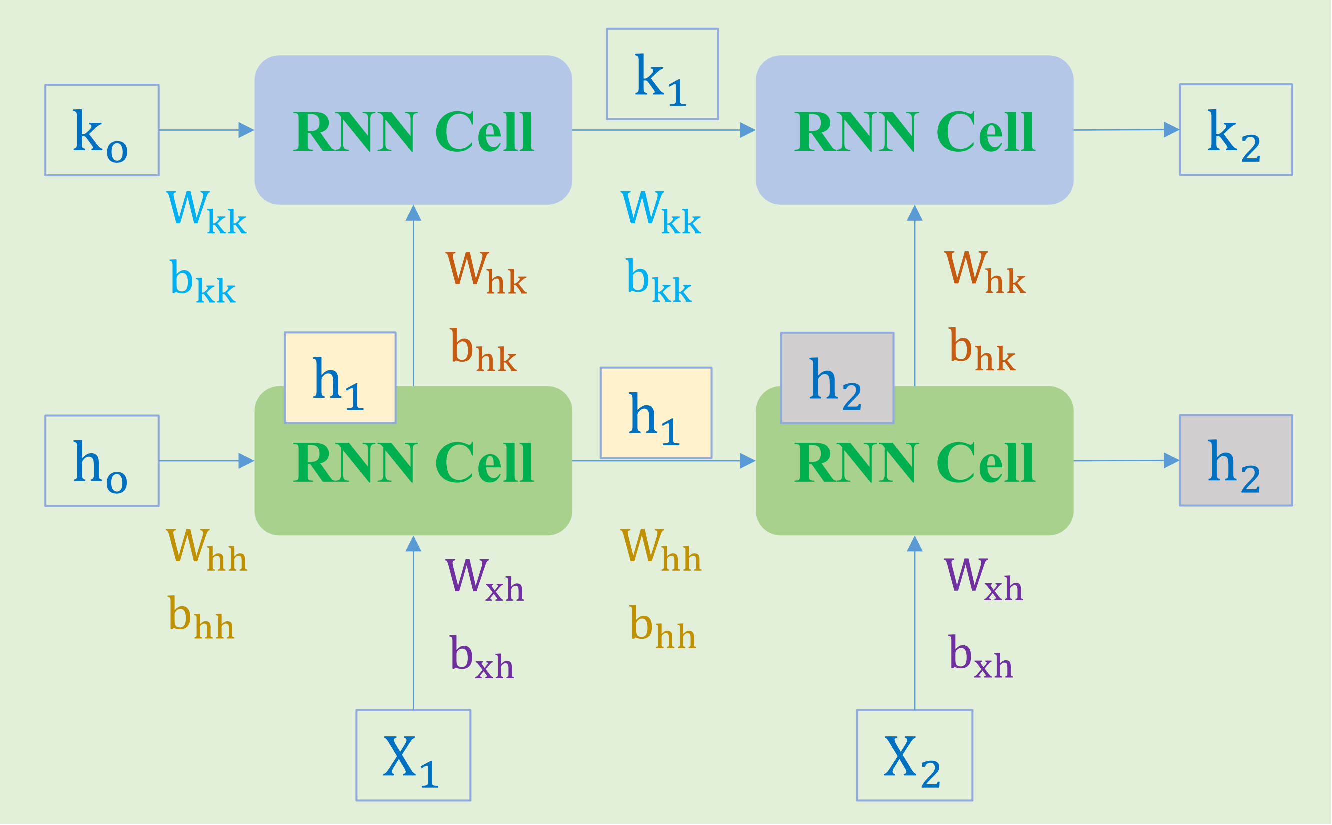
$$k_1 = \tanh(W_{hk}h_1 + b_{hk} + W_{kk}k_0 + b_{kk})$$

$$k_2 = \tanh(W_{hk}h_2 + b_{hk} + W_{kk}k_1 + b_{kk})$$

$$h_1 = \tanh(W_{xh}X_1 + b_{xh} + W_{hh}h_0 + b_{hh})$$

$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh})$$





$$\mathbf{W_{hh}} = \begin{bmatrix} 0.9 & -0.1 & 0.1 \\ -0.1 & -0.9 & 0.3 \\ 0.1 & -0.3 & -0.9 \end{bmatrix} \qquad \mathbf{W_{kk}} = \begin{bmatrix} 0.1 & 0.9 \\ 0.9 & -0.1 \end{bmatrix}$$

$$\mathbf{W_{xh}} = \begin{bmatrix} -0.2 & -0.4 & 0.3 & -0.4 \\ -0.6 & -0.8 & 0.5 & -0.3 \\ 0.1 & -0.1 & 0.6 & 0.1 \end{bmatrix} \qquad \mathbf{W_{hk}} = \begin{bmatrix} -1.1 & 0.3 & -0.1 \\ 0.1 & -0.2 & 0.4 \end{bmatrix}$$

$$\mathbf{b_{xh}} = \mathbf{b_{hh}} = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} \qquad \mathbf{b_{hk}} = \mathbf{b_{kk}} = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$k_1 = \tanh(W_{hk}h_1 + b_{hk} + W_{kk}k_0 + b_{kk}) = \begin{bmatrix} 0.49 \\ 0.407 \end{bmatrix}$$

$$k_2 = \tanh(W_{hk}h_2 + b_{hk} + W_{kk}k_1 + b_{kk}) = \begin{bmatrix} 0.84 \\ 0.42 \end{bmatrix}$$

$$h_1 = \tanh(W_{xh}X_1 + b_{xh} + W_{hh}h_0 + b_{hh}) = \begin{bmatrix} -0.833 \\ -0.98 \\ 0.8 \end{bmatrix}$$

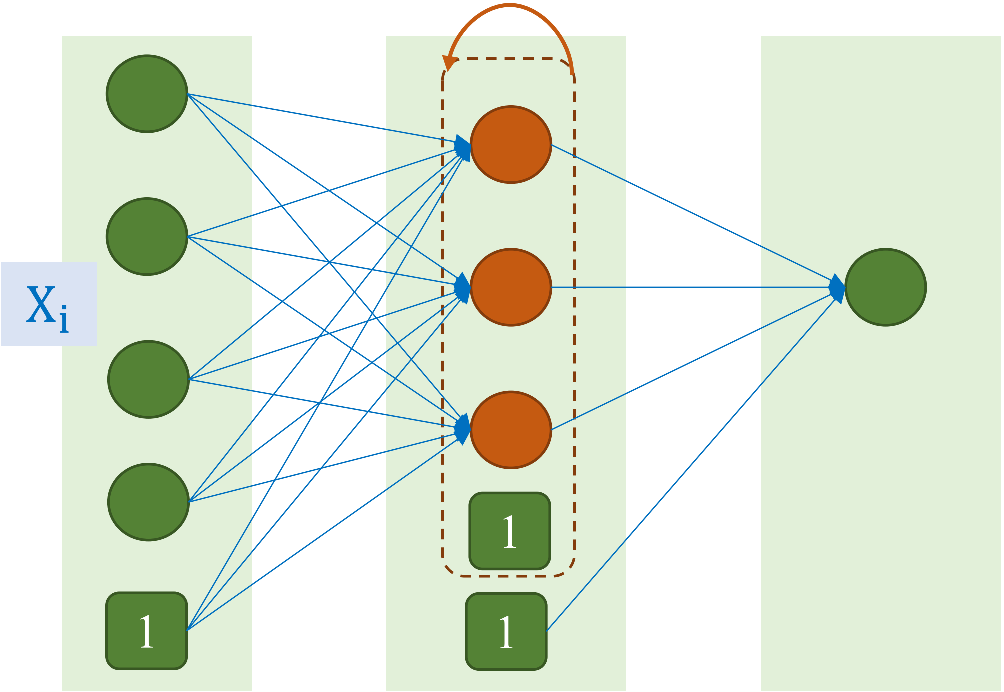
$$h_2 = \tanh(W_{xh}X_2 + b_{xh} + W_{hh}h_1 + b_{hh}) = \begin{bmatrix} -0.99 \\ -0.97 \\ -0.109 \end{bmatrix}$$

Text Deep Models

❖ Recurrent Neural Networks (RNN)

$$\mathbf{W}_{hh} = \begin{bmatrix} 0.226 & -0.068 & -0.971 \\ -0.973 & -0.014 & -0.226 \\ -0.001 & -0.997 & 0.070 \end{bmatrix} \qquad \mathbf{b}_{xh} = \mathbf{b}_{hh} = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

$$\mathbf{W}_{xh} = \begin{bmatrix} -0.436 & 0.373 & -0.032 & 0.403 \\ -0.452 & 0.296 & -0.609 & -0.643 \\ -0.761 & 0.266 & -0.526 & -0.703 \end{bmatrix}$$

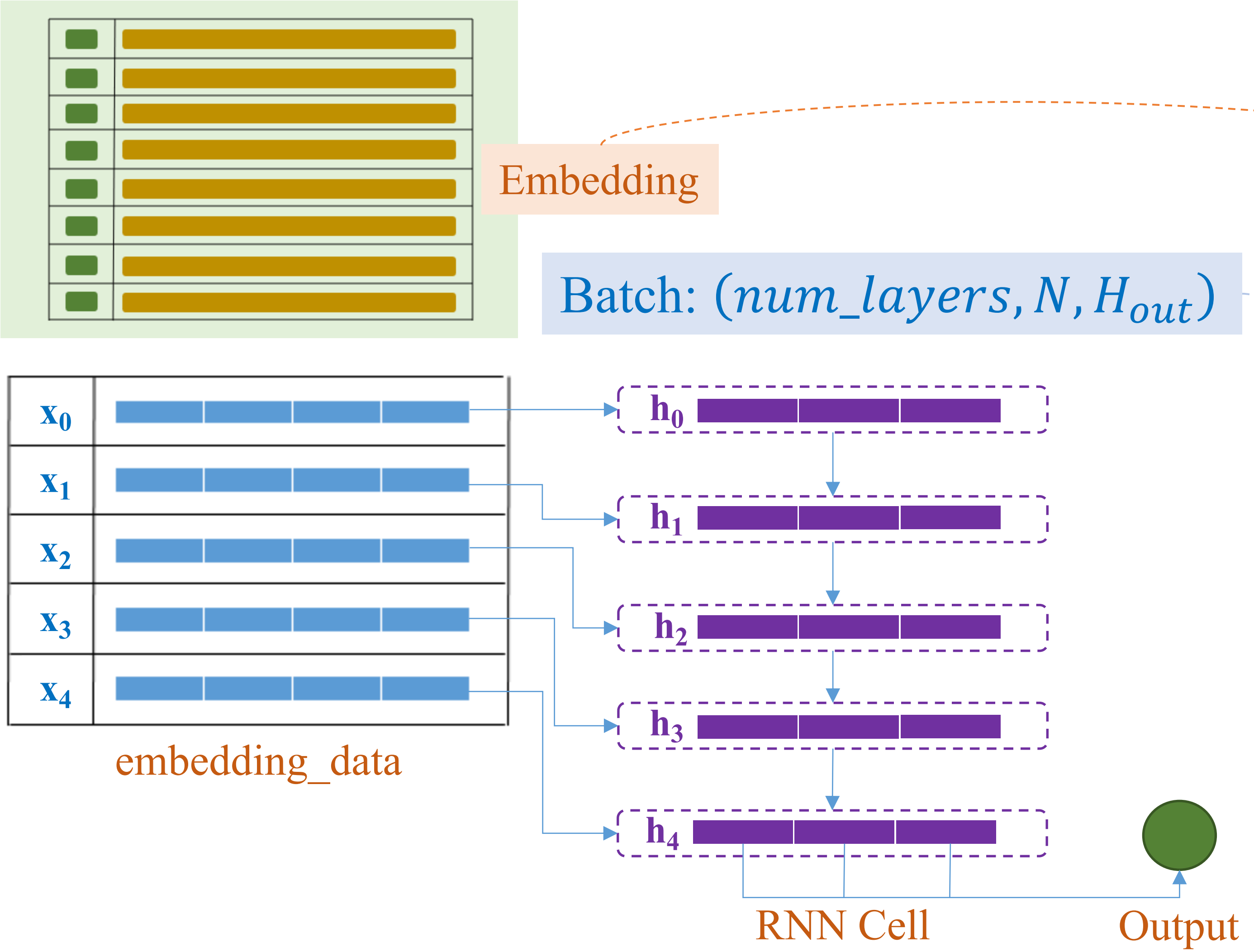



```python
class TextClsModel(nn.Module):
    def __init__(self, vocab_size=8,
                 emb_dim=4, hidden_dim=3):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size,
                                      emb_dim)
        self.rnn = nn.RNN(emb_dim, hidden_dim,
                          batch_first=True)
        self.fc = nn.Linear(hidden_dim, 1)

    def forward(self, x):
        x = self.embedding(x)
        _, h_rnn = self.rnn(x)
        last_h_rnn = h_rnn[-1,:,:]
        x = self.fc(last_h_rnn)
        return x
```

$$\mathbf{h}_t = \tanh(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_{hh} + \mathbf{W}_{xh}\mathbf{x}_t + \mathbf{b}_{xh})$$

$$\hat{y} = \text{sigmoid}(\mathbf{W}_D\mathbf{h}_t + \mathbf{b}_D)$$



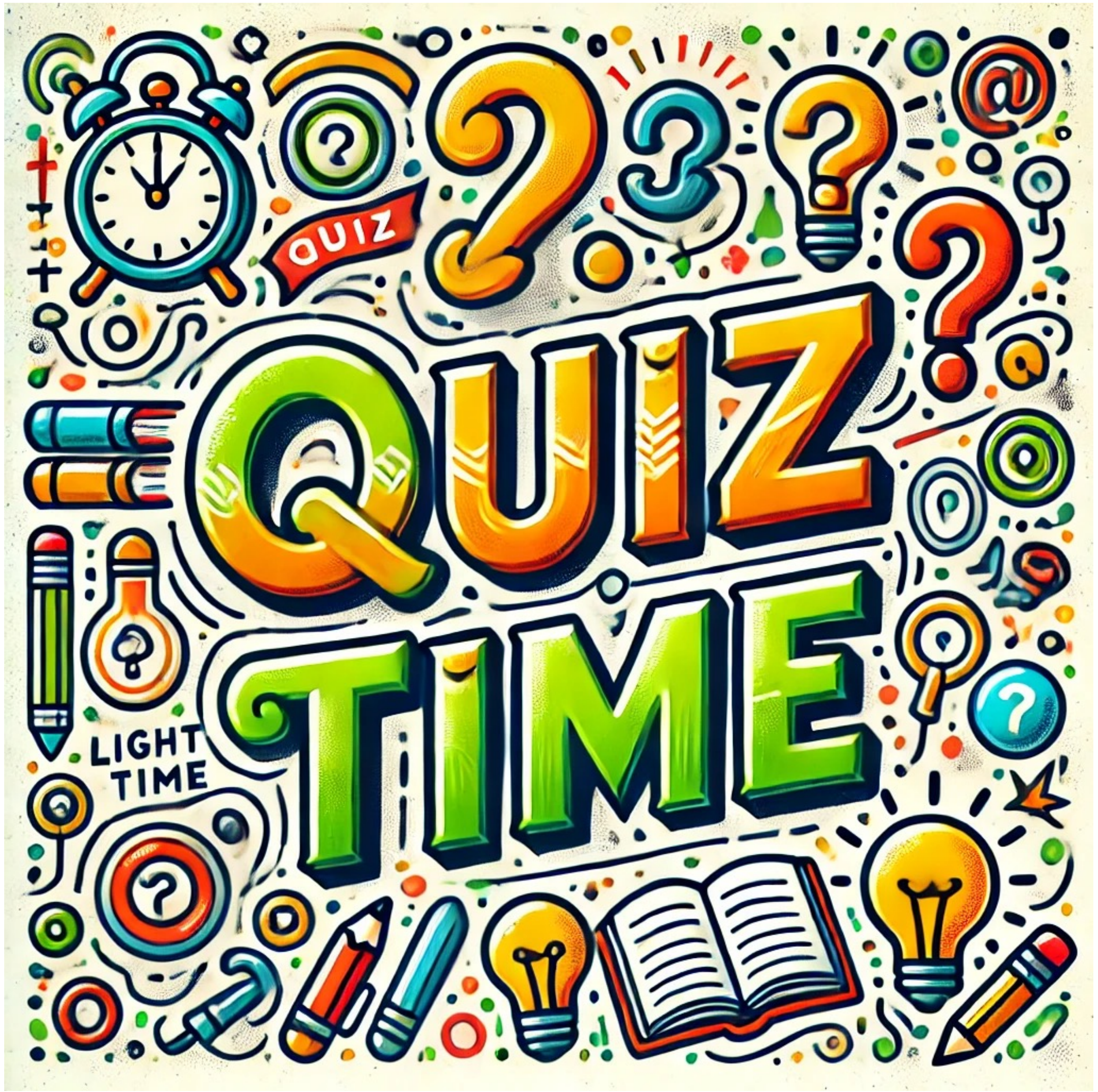

```
class TextClsModel(nn.Module):
    def __init__(self, vocab_size=8,
                 emb_dim=4, hidden_dim=3):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size,
                                      emb_dim)
        self.rnn = nn.RNN(emb_dim, hidden_dim,
                          batch_first=True)
        self.fc = nn.Linear(hidden_dim, 1)

    def forward(self, x):
        x = self.embedding(x)
        _, h_rnn = self.rnn(x)
        last_h_rnn = h_rnn[-1,:,:]
        x = self.fc(last_h_rnn)
        return x

model = TextClsModel(vocab_size=8,
                     emb_dim=4,
                     hidden_dim=3)
```

```
from torchsummary import summary
seq_length = 5
random_tensor = torch.randint(low=0, high=8,
                              size=(64, seq_length),
                              dtype=torch.long)
summary(model, random_tensor)
```

```
==========================================================================
Layer (type:depth-idx)                  Output Shape              Param
==========================================================================
├─Embedding: 1-1                        [-1, 5, 4]                32
├─RNN: 1-2                              [-1, 5, 3]                27
├─Linear: 1-3                           [-1, 1]                   4
==========================================================================
```


QUIZ
QUIZ
TIME
LIGHT
TIME

Outline

SECTION 1

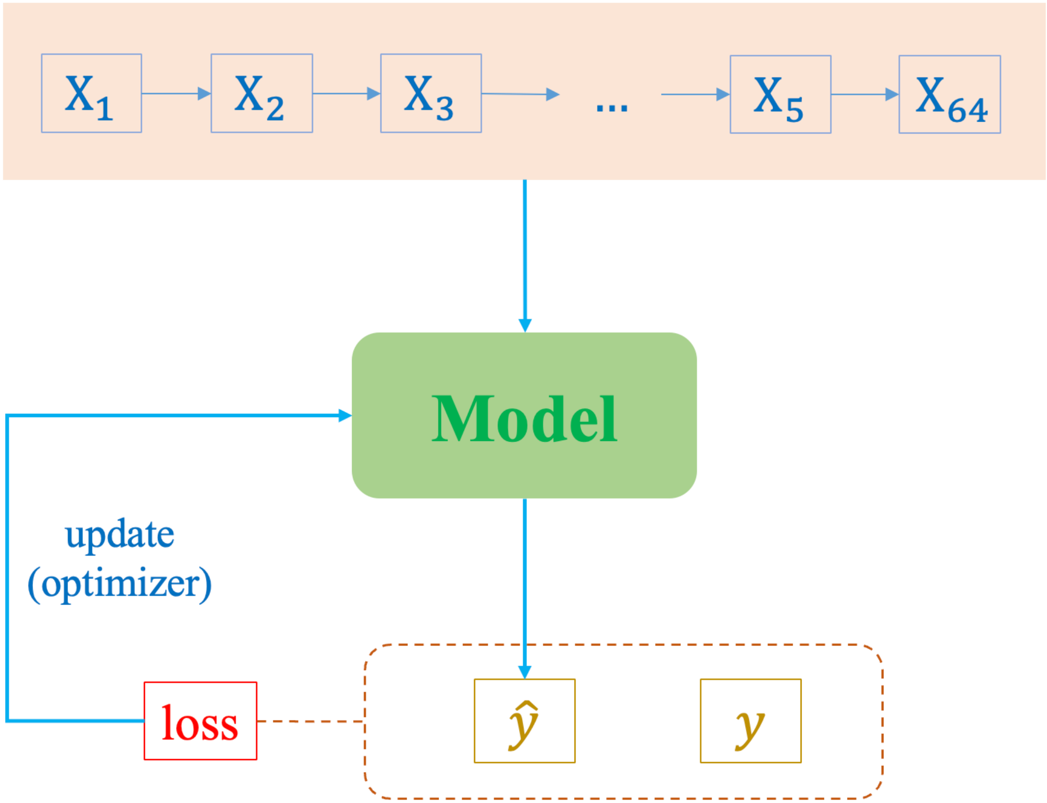
Text Classification

SECTION 2

RNN Construction

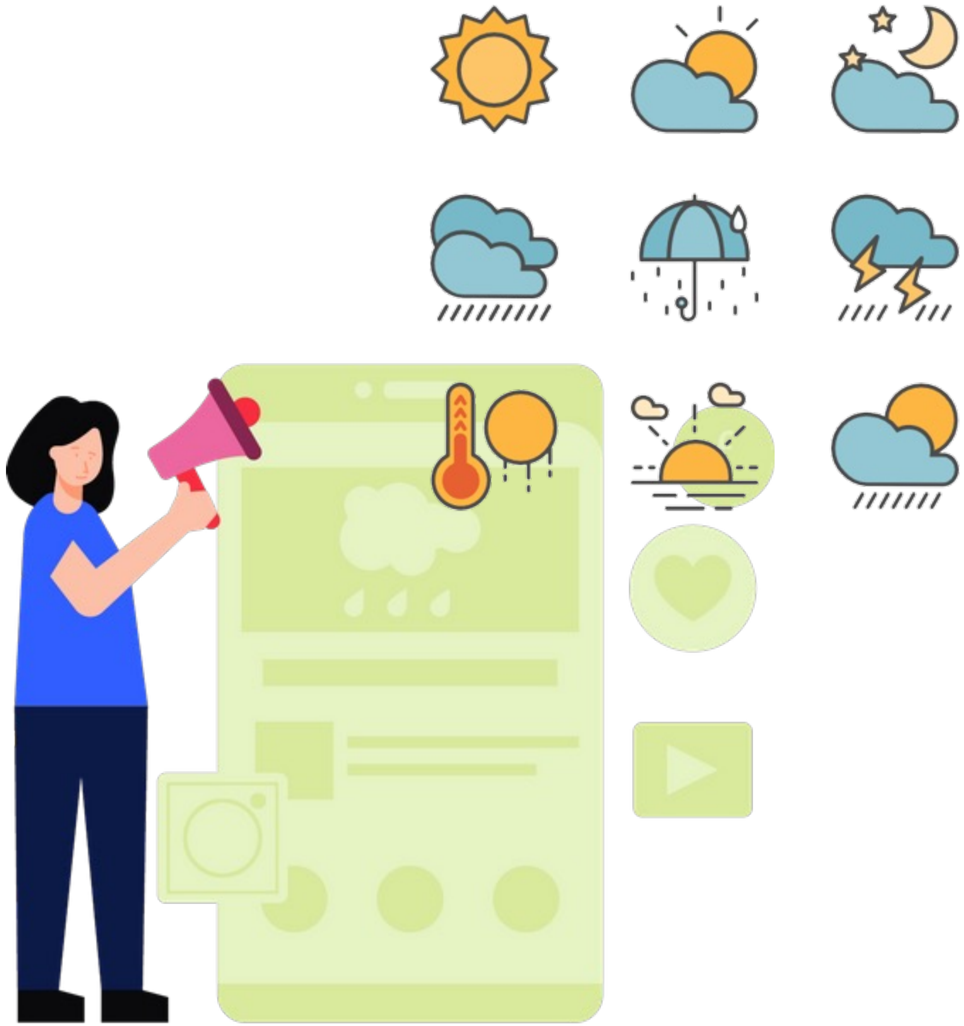
SECTION 3

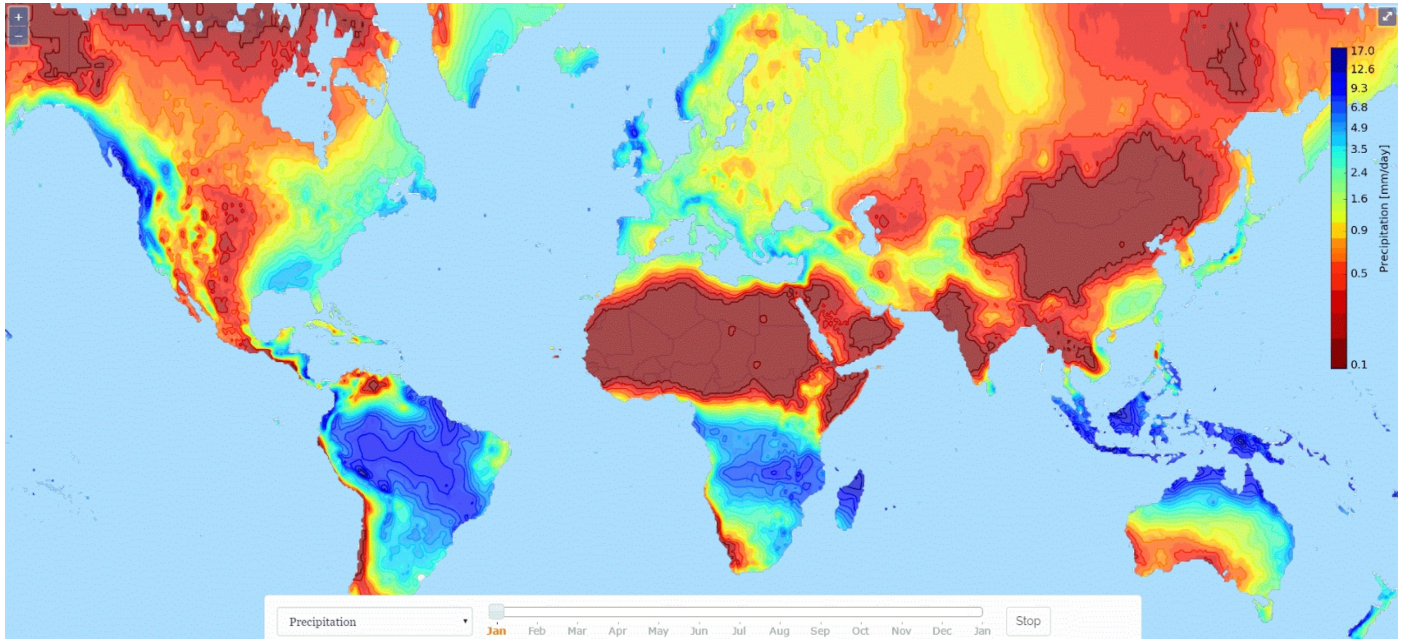
RNNs for Time-series



Weather Forecasting

❖ Introduction





Predict future temperature in weather forecasting

Weather Forecasting

❖ Introduction

Problem Statement: Given temperature from the previous 5 hours (including the current one), predict temperature of the next 1 hour.

Hour	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00
Condition								
Temperature	32	31	31	30	29	26	25	

Time-series Data

Temperature forecasting

Date	Summary	Precip Type	Temperature (C)	Apparent Temperature (C)	Humidity	Wind Speed (km/h)	Wind Bearing (degrees)	Visibility (km)
2006-04-01 00	Partly Cloudy	rain	9.472222222	7.388888889	0.89	14.1197	251	15.8263
2006-04-01 01	Partly Cloudy	rain	9.355555556	7.227777778	0.86	14.2646	259	15.8263
2006-04-01 02	Mostly Cloudy	rain	9.377777778	9.377777778	0.89	3.9284	204	14.9569
2006-04-01 03	Partly Cloudy	rain	8.288888889	5.944444444	0.83	14.1036	269	15.8263
2006-04-01 04	Mostly Cloudy	rain	8.755555556	6.977777778	0.83	11.0446	259	15.8263
2006-04-01 05	Partly Cloudy	rain	9.222222222	7.111111111	0.85	13.9587	258	14.9569
2006-04-01 06	Partly Cloudy	rain	7.733333333	5.522222222	0.95	12.3648	259	9.982
2006-04-01 07	Partly Cloudy	rain	8.772222222	6.527777778	0.89	14.1519	260	9.982
2006-04-01 08	Partly Cloudy	rain	10.82222222	10.82222222	0.82	11.3183	259	9.982
2006-04-01 09	Partly Cloudy	rain	13.77222222	13.77222222	0.72	12.5258	279	9.982
2006-04-01 10	Partly Cloudy	rain	16.01666667	16.01666667	0.67	17.5651	290	11.2056
2006-04-01 11	Partly Cloudy	rain	17.14444444	17.14444444	0.54	19.7869	316	11.4471
2006-04-01 12	Partly Cloudy	rain	17.8	17.8	0.55	21.9443	281	11.27
2006-04-01 13	Partly Cloudy	rain	17.33333333	17.33333333	0.51	20.6885	289	11.27
2006-04-01 14	Partly Cloudy	rain	18.87777778	18.87777778	0.47	15.3755	262	11.4471
2006-04-01 15	Partly Cloudy	rain	18.91111111	18.91111111	0.46	10.4006	288	11.27
2006-04-01 16	Partly Cloudy	rain	15.38888889	15.38888889	0.6	14.4095	251	11.27
2006-04-01 17	Mostly Cloudy	rain	15.55	15.55	0.63	11.1573	230	11.4471
2006-04-01 18	Mostly Cloudy	rain	14.25555556	14.25555556	0.69	8.5169	163	11.2056
2006-04-01 19	Mostly Cloudy	rain	13.14444444	13.14444444	0.7	7.6314	139	11.2056
2006-04-01 20	Mostly Cloudy	rain	11.55	11.55	0.77	7.3899	147	11.0285
2006-04-01 21	Mostly Cloudy	rain	11.18333333	11.18333333	0.76	4.9266	160	9.982
2006-04-01 22	Partly Cloudy	rain	10.11666667	10.11666667	0.79	6.6493	163	15.8263
2006-04-01 23	Mostly Cloudy	rain	10.2	10.2	0.77	3.9284	152	14.9569
2006-04-10 00	Partly Cloudy	rain	10.42222222	10.42222222	0.62	16.9855	150	15.8263
2006-04-10 01	Partly Cloudy	rain	9.911111111	7.566666667	0.66	17.2109	149	15.8263
2006-04-10 02	Mostly Cloudy	rain	11.18333333	11.18333333	0.8	10.8192	163	14.9569
2006-04-10 03	Partly Cloudy	rain	7.155555556	5.044444444	0.79	11.0768	180	15.8263
2006-04-10 04	Partly Cloudy	rain	6.111111111	4.816666667	0.82	6.6493	161	15.8263

Weather Forecasting

❖ Introduction

Time	Temperature (C)
2006-04-01 00:00:00.000 +0200	9.472222
2006-04-01 01:00:00.000 +0200	9.355556
2006-04-01 02:00:00.000 +0200	9.377778
2006-04-01 03:00:00.000 +0200	8.288889
2006-04-01 04:00:00.000 +0200	8.755556
2006-04-01 05:00:00.000 +0200	9.222222

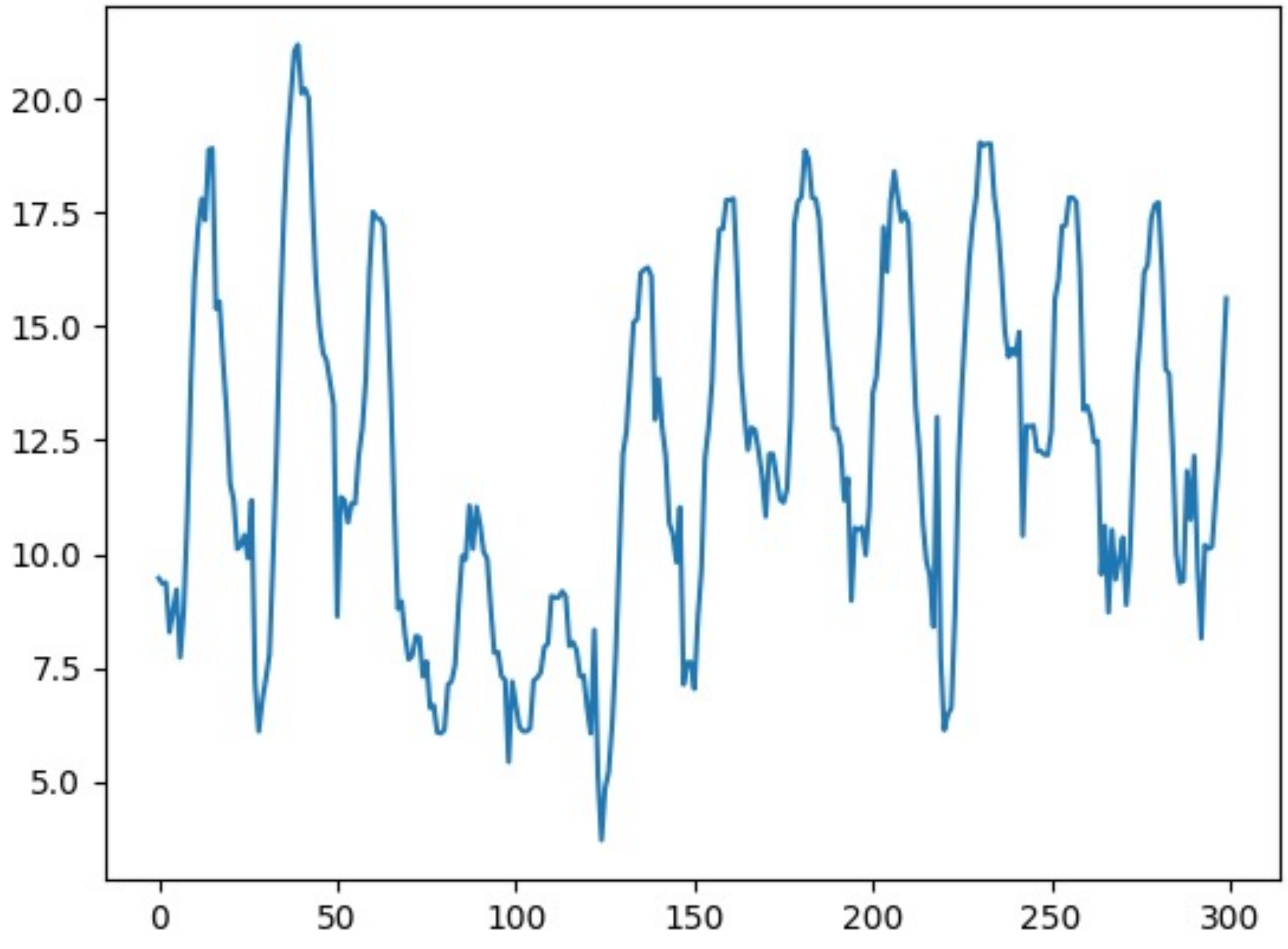
X

y

Temperature forecasting datatable

Time-series Data

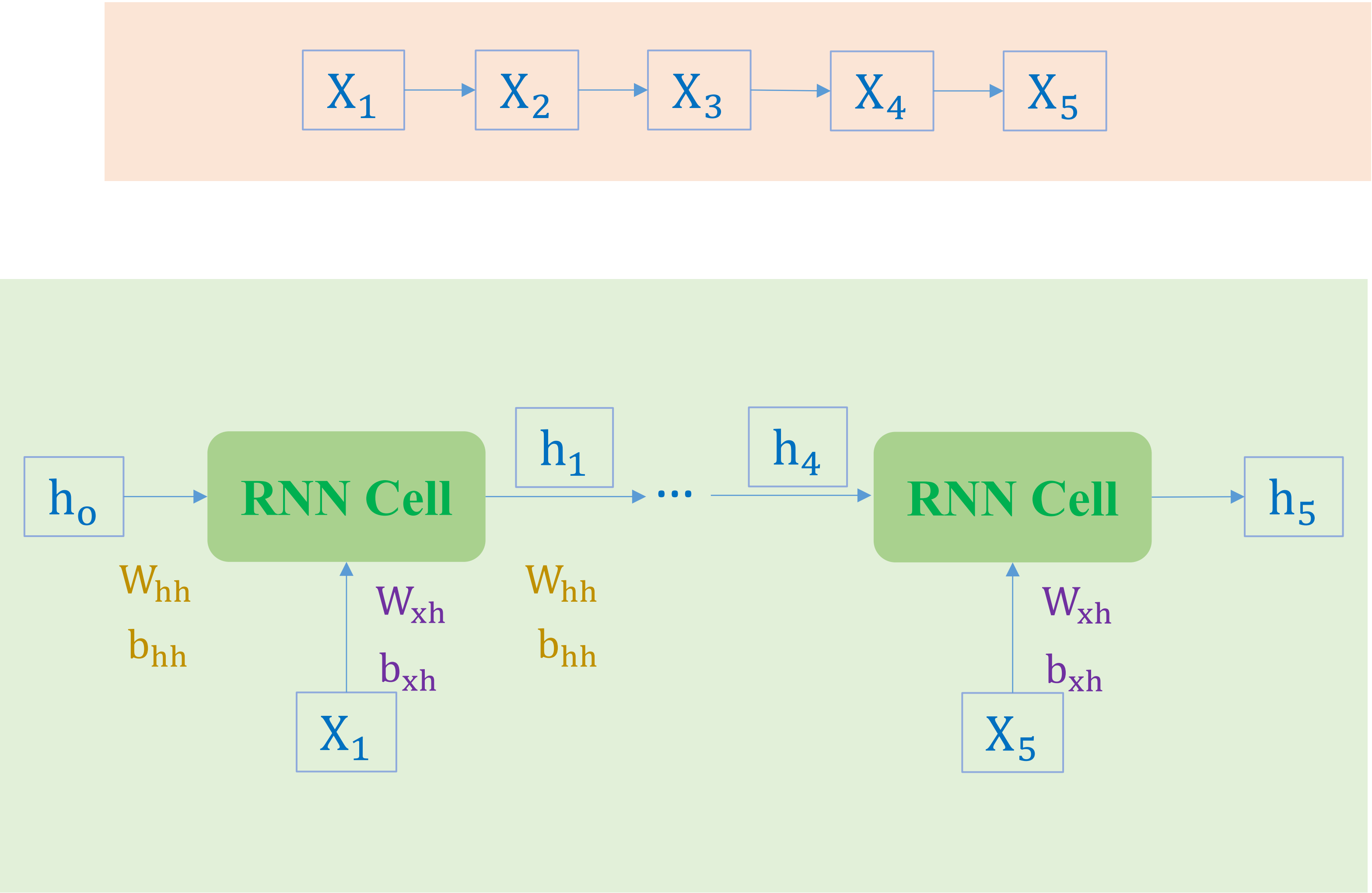
❖ Temperature forecasting

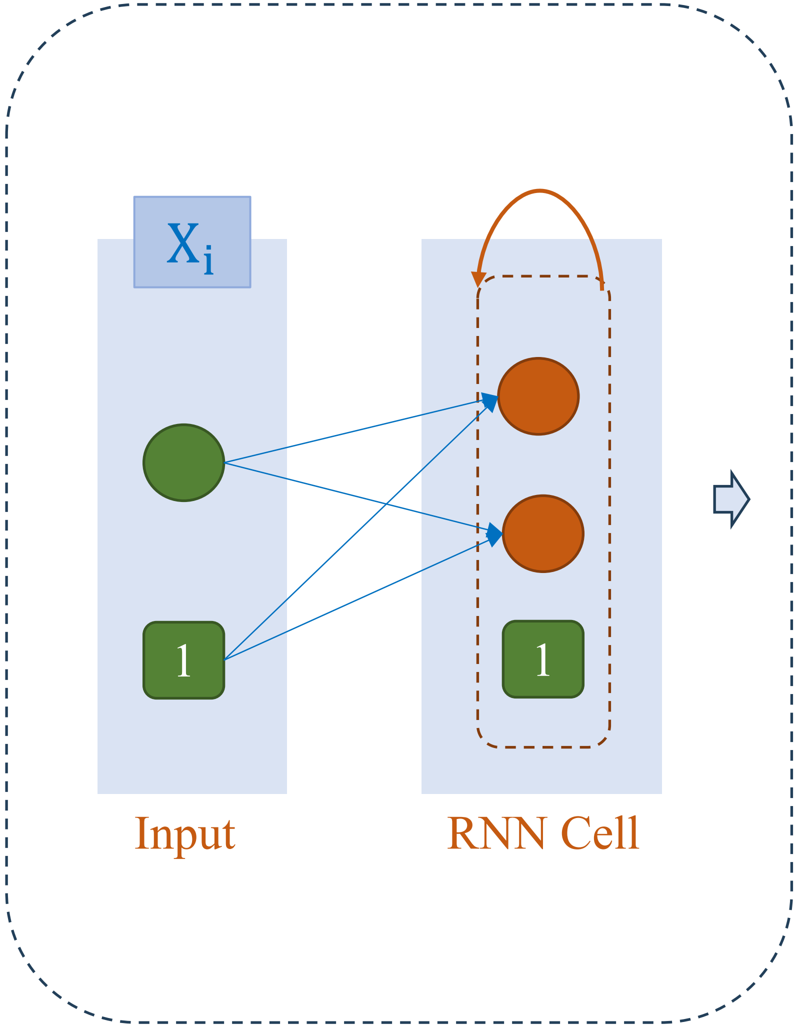


Date	Temperature (C)
2006-04-01 00	9.472222222
2006-04-01 01	9.355555556
2006-04-01 02	9.377777778
2006-04-01 03	8.288888889
2006-04-01 04	8.755555556
2006-04-01 05	9.222222222
2006-04-01 06	7.733333333
2006-04-01 07	8.772222222
2006-04-01 08	10.82222222
2006-04-01 09	13.77222222
2006-04-01 10	16.01666667
2006-04-01 11	17.14444444
2006-04-01 12	17.8
2006-04-01 13	17.33333333
2006-04-01 14	18.87777778
2006-04-01 15	18.91111111
2006-04-01 16	15.38888889
2006-04-01 17	15.55
2006-04-01 18	14.25555556
2006-04-01 19	13.14444444
2006-04-01 20	11.55
2006-04-01 21	11.18333333
2006-04-01 22	10.11666667
2006-04-01 23	10.2

Time-series Data

❖ Temperature forecasting





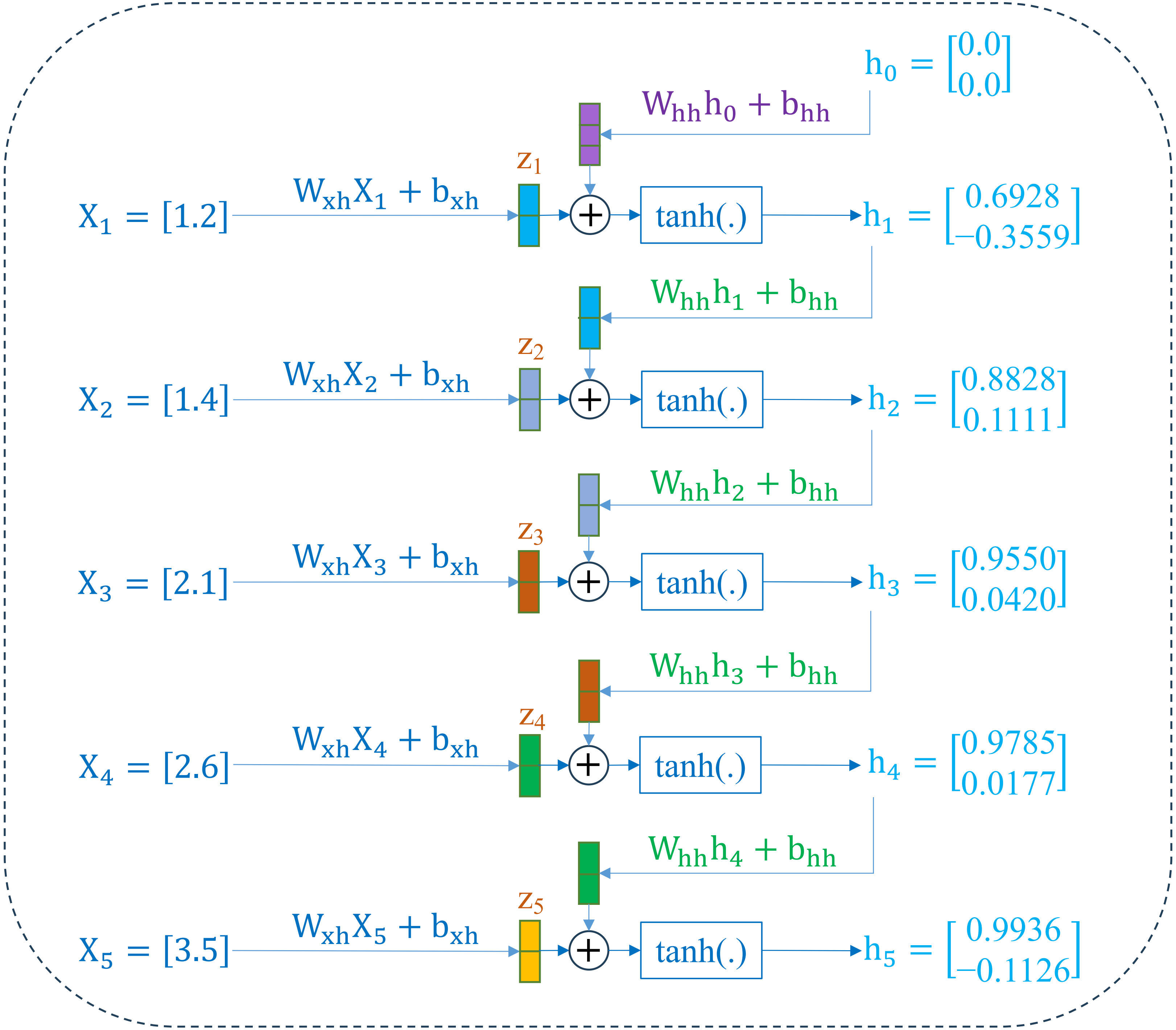
Example

$$W_{xh} = \begin{bmatrix} 0.6584 \\ -0.1671 \end{bmatrix}$$

$$b_{xh} = \begin{bmatrix} -0.5966 \\ 0.0945 \end{bmatrix}$$

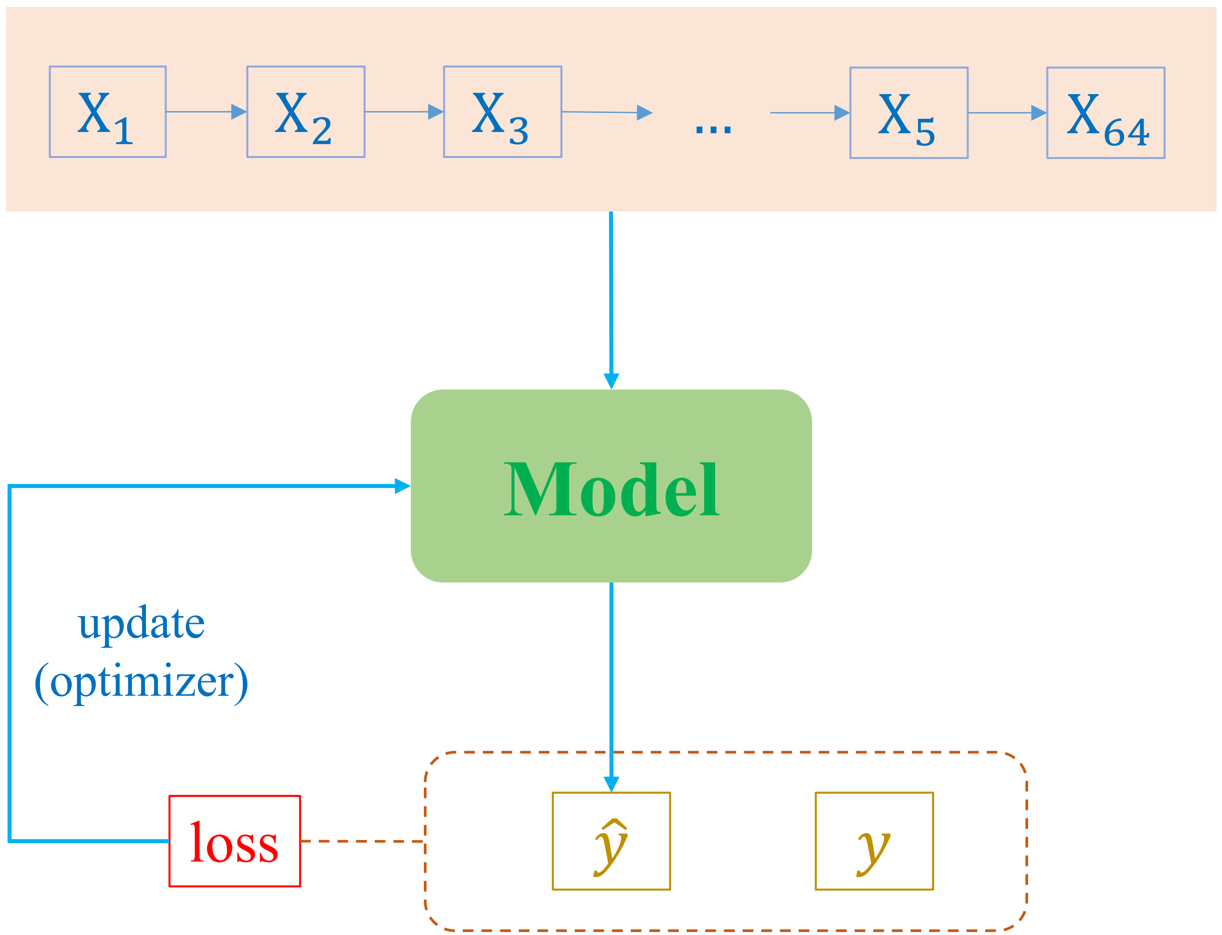
$$W_{hh} = \begin{bmatrix} 0.5147 & -0.1310 \\ 0.6606 & -0.1671 \end{bmatrix}$$

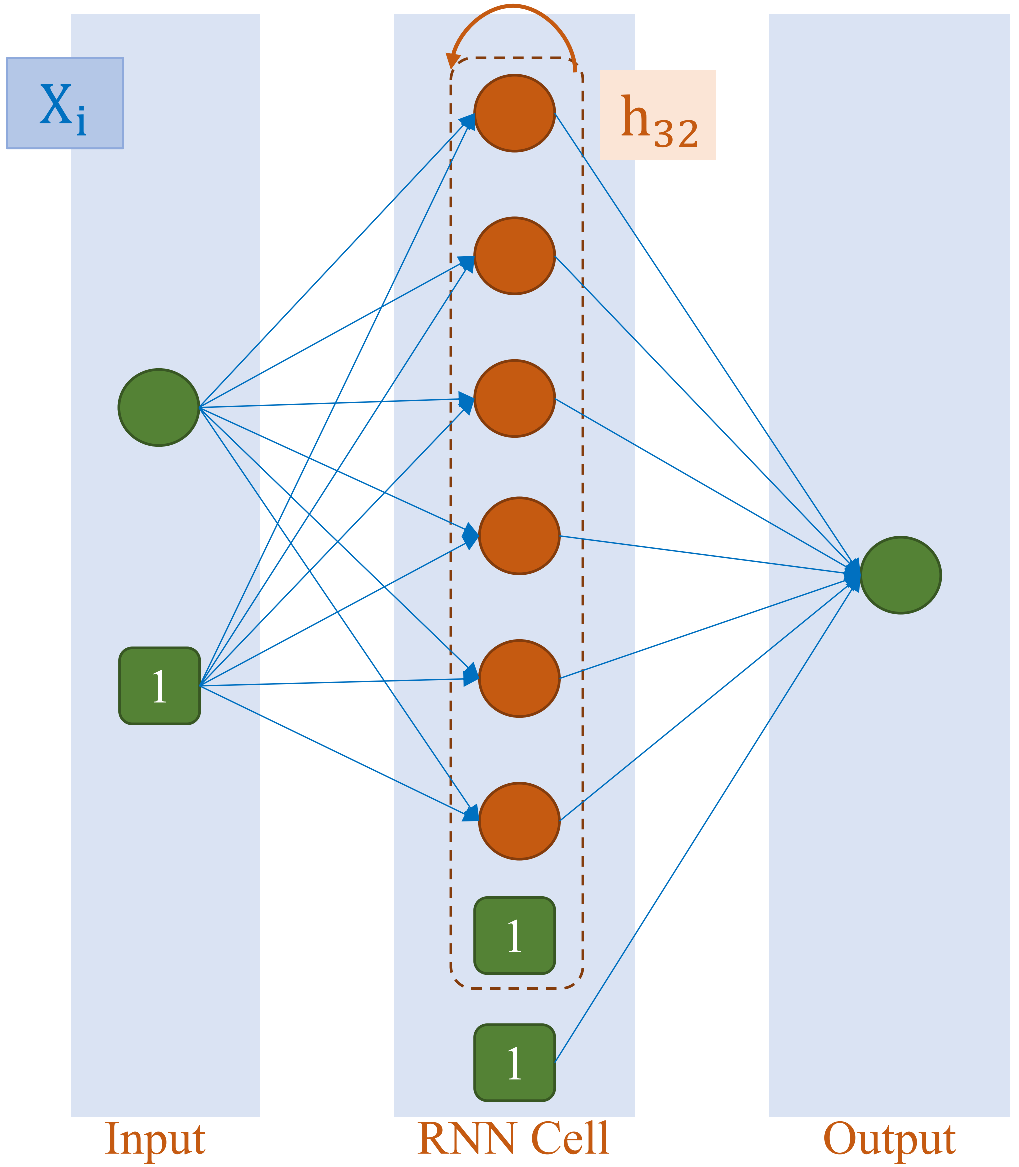
$$b_{hh} = \begin{bmatrix} 0.6599 \\ -0.2662 \end{bmatrix}$$



Implementation

❖ Temperature forecasting





Implementation

❖ Back to Temperature forecasting

sequence_length = 64   embed_dim = 1

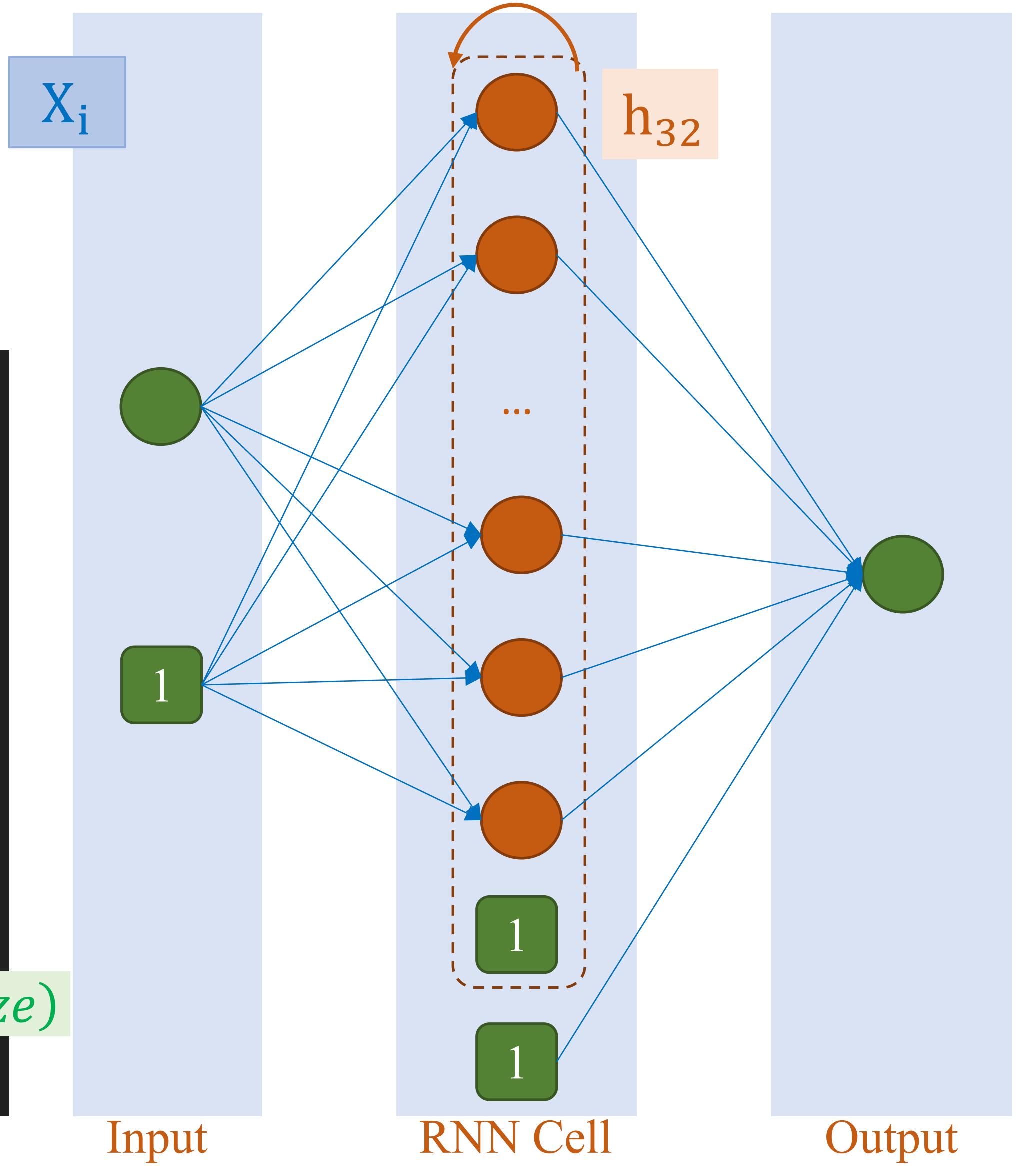
output_dim = 1   hidden_dim = 32

```
class RNNModel(nn.Module):
    def __init__(self, hidden_dim, output_dim):
        super(RNNModel, self).__init__()
        self.rnn = nn.RNN(1, hidden_dim, batch_first=True)
        self.fc = nn.Linear(hidden_dim, output_dim)

    def forward(self, x):
        output_rnn, hidden_rnn = self.rnn(x)
        last_hidden = hidden_rnn[-1,:,:]
        output = self.fc(last_hidden)
        return output

model = RNNModel(hidden_dim=32, output_dim=1)
```

$(num\_rnn\_layers, N, hidden\_size)$



Implementation

❖ Stack of three RNNs

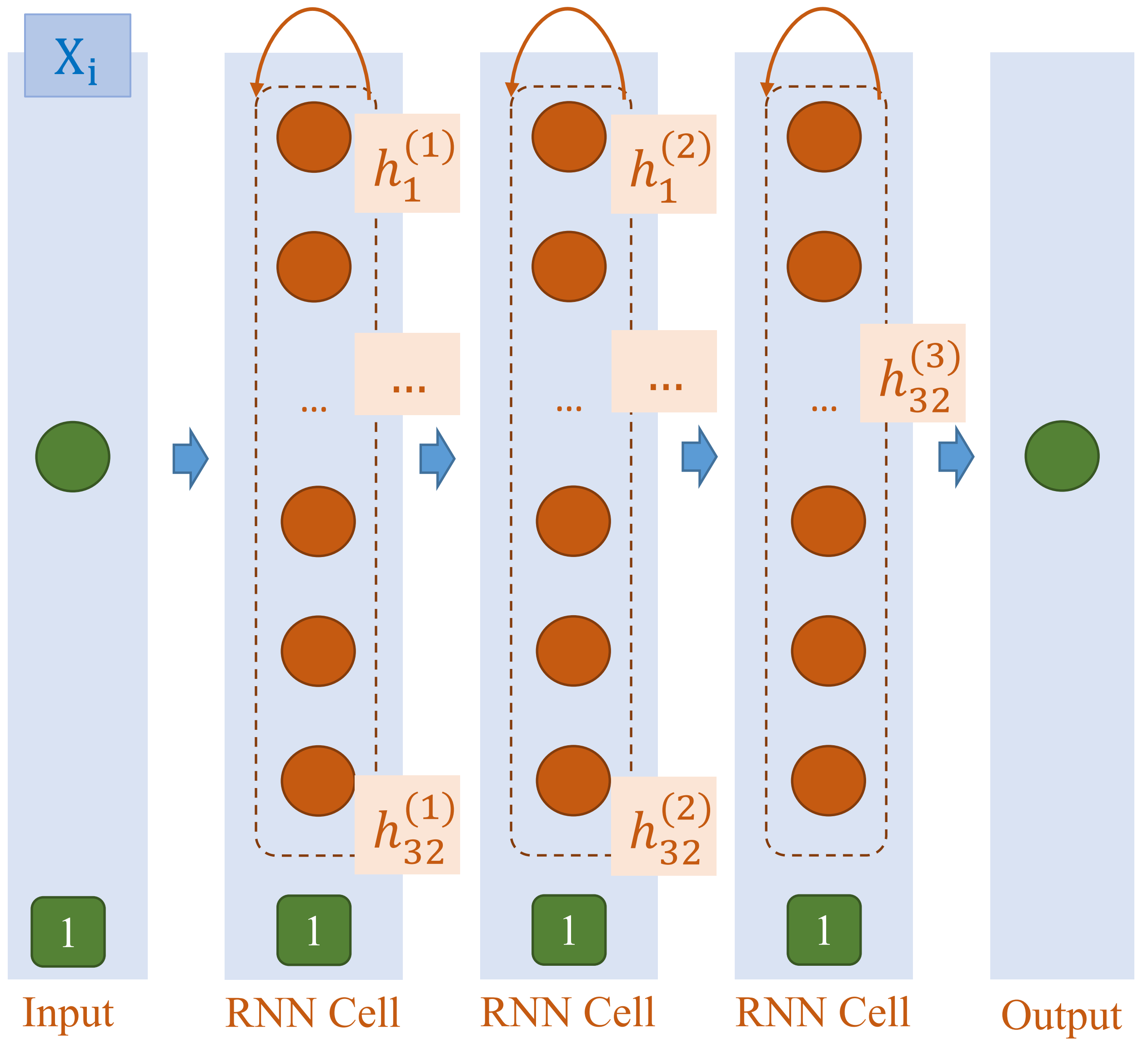
sequence_length = 64   embed_dim = 1

output_dim = 1   hidden_dim = 32

```
class RNNModel(nn.Module):
    def __init__(self, hidden_dim, output_dim):
        super(RNNModel, self).__init__()
        self.rnn = nn.RNN(1, hidden_dim,
                          num_layers=3,
                          batch_first=True)
        self.fc = nn.Linear(hidden_dim, output_dim)

    def forward(self, x):
        output_rnn, hidden_rnn = self.rnn(x)
        last_hidden = hidden_rnn[-1,:,:]
        output = self.fc(last_hidden)
        return output

model = RNNModel(hidden_dim=32, output_dim=1)
```



Implementation

❖ Data preparation

```python
def prepare_data(data, lag, ahead, train_ratio, batch_size):
    # Create sequences
    X, y = create_sequences(data, lag, ahead)

    # Flatten all the features of a sample for RNN
    X = X.reshape(X.shape[0], -1, 1)

    # Split the data
    train_size = int(len(X) * train_ratio)
    X_train, X_test = X[:train_size], X[train_size:]
    y_train, y_test = y[:train_size], y[train_size:]

    # Convert to PyTorch tensors
    X_train_tensor = torch.tensor(X_train, dtype=torch.float32)
    y_train_tensor = torch.tensor(y_train, dtype=torch.float32)
    X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
    y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

    # ...
```

(N, L, H_{in}) when `batch_first=True`

```python
def create_sequences(data, lag, ahead):
    X, y = [], []
    for i in range(len(data) - lag - ahead + 1):
        X.append(data[i:(i + lag)])
        y.append(data[(i + lag):(i + lag + ahead)])
    return np.array(X), np.array(y)
```

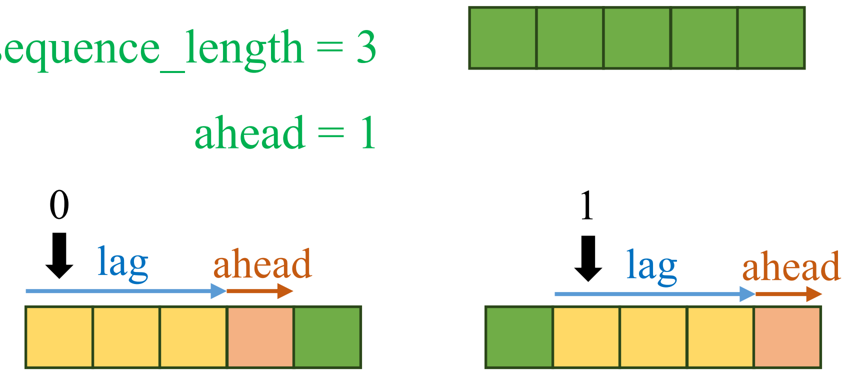
```python
X, y = create_sequences(data, lag, ahead)
print(X.shape)
print(y.shape)
```

```
(96389, 64)
(96389, 1)
```

train_data_length = 5

lag = sequence_length = 3

ahead = 1



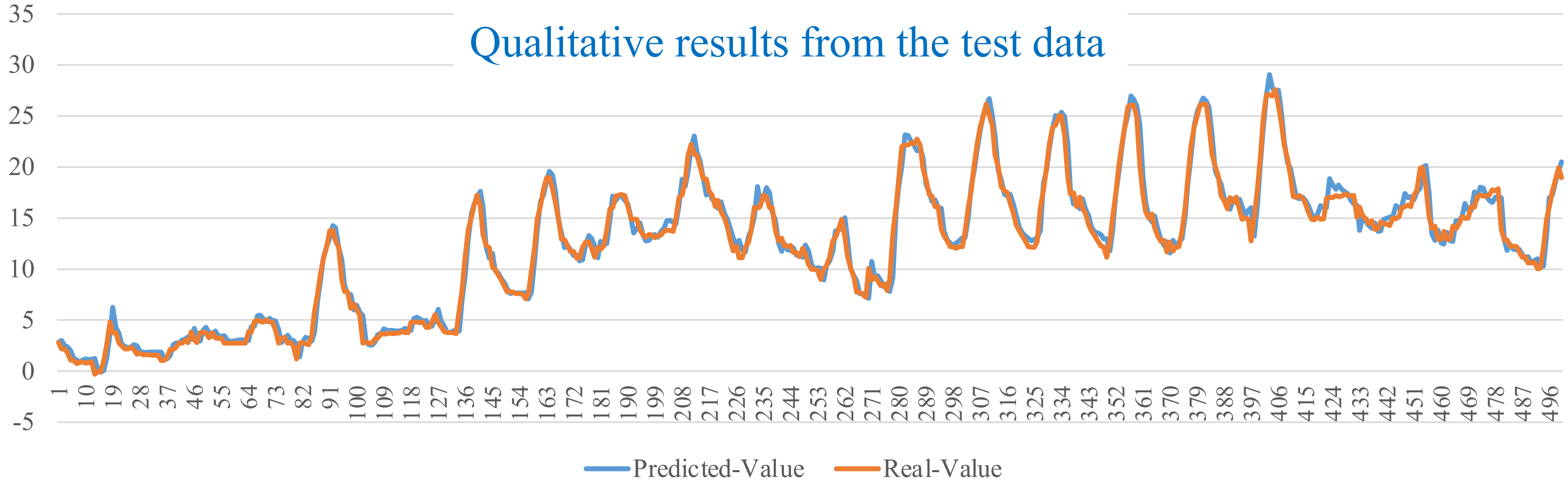
range(5-3-1+1) = range(2) → 0, 1

Implementation

Train

```python
criterion = nn.MSELoss()
optimizer = Adam(model.parameters(),
                 lr=lr)
```

```python
def train_model(model, criterion, optimizer, train_loader, num_epochs):
    for epoch in range(num_epochs):
        model.train()
        for i, (sequences, labels) in enumerate(train_loader):
            sequences, labels = sequences.to(device), labels.to(device)
            # Forward pass
            outputs = model(sequences)
            loss = criterion(outputs, labels)
            # Backward and optimize
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()
    return model, losses
```



R2 Score: 0.984
MAE: 0.71
MSE: 1.23

Common Metrics for TS Data

y	11.18	11.66	8.97
$\hat{y}$	-0.33	-0.61	1.35
$y-\hat{y}$	11.51	12.27	7.62
$(y-\hat{y})^2$	132.48	150.55	50.06

$$\bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i$$

$$\bar{y} = \frac{1}{3}(11.18 + 11.66 + 8.97)$$
$$= 10.6$$

$y-\bar{y}$	0.58	1.06	-1.63
$(y-\bar{y})^2$	0.336	1.123	2.656

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{3}(132.48 + 150.55 + 50.06)$$
$$= 111.03$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

$$MAE = \frac{1}{3}(|11.51| + |12.27| + |7.62|)$$
$$= 10.46$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_i)^2}$$

$$R^2 = 1 - \frac{132.48 + 150.55 + 50.06}{0.3364 + 1.123 + 2.6569}$$
$$= -79.91$$

Objectives

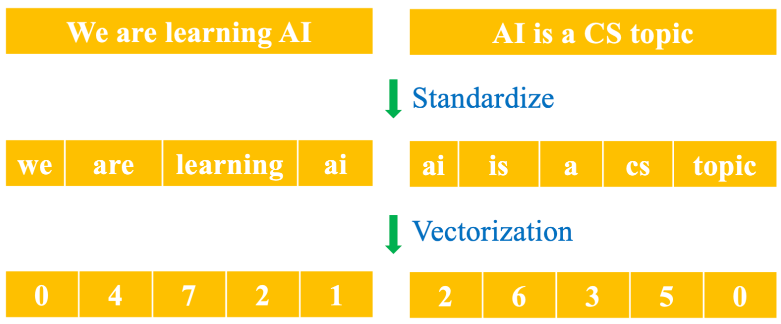
Text Classification

sample1: ‘We are learning AI’

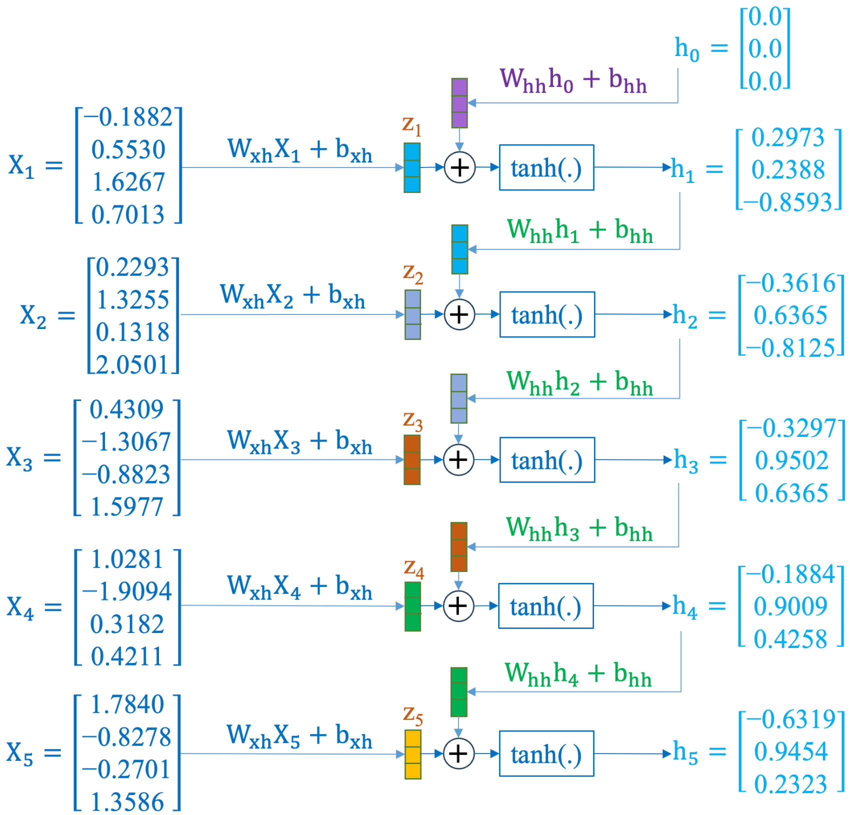
sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

(2) Transform text into features



RNN Construction



RNNs for Time-series